MANIFESTO

"Treatment-Resistant Schizophrenia": A Diagnosis Without a Measurement — Pseudo-Resistance, Dopamine Receptor Supersensitivity, and the Iatrogenic Revolving Door in Australian Psychiatry

Submission in Support of a Royal Commission into Antipsychotic Prescribing, Monitoring, Withdrawal and De-prescribing Protocols in NSW Public Mental Health Services

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Status: Public record. May be cited with attribution.

DISCLAIMER: This document advances a systemic argument, evidenced by one fully documented case. It is not medical advice, not a prescription, and not a treatment recommendation for any person. It records events, clinical findings, and published literature. Clinical figures are cited to their sources; where a figure is an estimate or a projection, it is labelled as such; where an input is not yet independently verified, it is flagged. The author is the appointed Enduring Guardian, NCAT Guardian and Financial Manager of the person whose case is described and holds lawful authority over his healthcare.

EVIDENTIARY BASIS AND THE EFFECT OF NON-PRODUCTION. The clinical record of the exemplar case is held exclusively by the health service and has been withheld across five statutory requests over eleven months. Accordingly, where this submission states that a test, protocol, assessment or result does not appear, that statement is made in the only form the evidence permits: **no such record has been produced to the patient or to his Guardian.** It is not an assertion about what the withheld record contains. That distinction is not the author's to resolve. A service that declines to produce a record cannot at the same time rely on its contents; and any document produced in answer to this submission will itself establish that the record existed and was withheld. Where the Guardian's own contemporaneous observation, video record or written record is the source, it is identified as such.

THE SHORT POINT

1. A diagnosis is being used as a disposal. **"Treatment-resistant schizophrenia" (TRS)** is applied to patients who were never shown to have the condition and never received the treatment they resisted. In a documented subset, the "resistance" is an artefact of pharmacology — the drug never reached a therapeutic concentration in the patient's body — and the "schizophrenia" is a drug-induced or withdrawal-driven state that the treatment itself perpetuates.

2. The engine of the revolving door is **dopamine receptor supersensitivity (DRS)**: chronic, tight D2-receptor blockade up-regulates receptor density and sensitivity, so that dose reduction, missed doses, absconding, or metabolic fluctuation produce a rebound psychosis worse than the original — which is then read as relapse of an intractable illness and answered with escalation. The label hardens; the door keeps turning.

3. This is compounded by three practices this submission asks a Royal Commission to examine: **rapid, high-percentage loading** of antipsychotics (titration measured in days, at percentage increases far above what published titration ceilings contemplate); **abrupt withdrawal** on absconding or discharge, with no inpatient taper protocols offered across all admissions; and **overmedication** — at least seven antipsychotic agents, poly-pharmacy, and dose escalation toward maximum in a patient the treating team's own genetic testing identified as unable to metabolise the drug to a therapeutic level.

4. The cost is not abstract. People carrying a schizophrenia diagnosis die **15–20 years** prematurely; for substance use disorder — the actual primary condition in the exemplar case — the gap is approximately **20 years** (Correll et al., *World Psychiatry* 2022; Chan et al., eClinicalMedicine 2023;65:102294)[1]. The literature identifies quality of care, antipsychotic use, and readmission frequency among the *modifiable* contributors to that gap. A system that manufactures resistance, escalates rather than tapers, and never treats cause is operating on the wrong side of every one of those modifiable factors.

5. The remedy is not the abolition of psychiatry. Acute psychiatric intervention is valid and necessary for life-threatening presentations. The remedy is accountability for the use of indefinite antipsychotic prescription — without pharmacogenomic testing, without tapering, without rehabilitation — as a substitute for treating trauma and substance use disorder, and for the systemic absence of the services that would make a different course possible.

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[1] Correll CU, Solmi M, Croatto G, Schneider LK, Rohani-Montez SC, Fairley L, Smith N, Bitter I, Gorwood P, Taipale H, Tiihonen J. *Mortality in people with schizophrenia: a systematic review and meta-analysis of relative risk and aggravating or attenuating factors*. World Psychiatry. 2022;21(2):248–271.

PART I. WHAT THE LABEL CONCEALS

A. "Treatment-Resistant Schizophrenia" — a conclusion masquerading as a finding

TRS is a term of art with strict criteria: a genuine schizophrenia diagnosis, and persistent symptoms **despite adequate trials** of at least two antipsychotics **at adequate dose, duration, and — critically — adequate plasma concentration**. Each element must be established, not assumed. In practice, the label is frequently reached by counting readmissions and drug switches, without confirming that any trial was ever pharmacologically adequate.

The point is conceded in Australian commentary on the RANZCP guidelines, which notes that to meet the international TRRIP criteria[2] two antipsychotics must have been tried at a dose and for a duration that should be effective, and that inter-individual variation in metabolism plays a part in whether that threshold was ever reached.

The distinction the literature draws is between true treatment resistance and **pseudo-resistance** — apparent non-response caused not by an intractable illness but by sub-therapeutic drug exposure (rapid metabolism, enzyme induction, non-adherence, or interaction). The two demand opposite responses. True resistance may warrant clozapine. **Pseudo-resistance does not: clozapine "does not appear to be the most adequate drug" where the failure is one of exposure rather than biology** (Cipolla et al., *Combination of Two Long-Acting Antipsychotics in Schizophrenia Spectrum Disorders: A Systematic Review*, *Brain Sciences*, 2024, as cited in HCCC HC00532615)[3]. Misclassifying pseudo-resistance as TRS routes a patient to the drug of last resort for a problem that drug cannot solve.

The label and the rationale do not match, and neither was ever established. "Treatment-resistant schizophrenia" appears in the record from at least 2025 — before the pharmacogenomic panel, before any plasma level, and before any trial had been shown to be pharmacologically adequate. But when the decision to commence clozapine was explained to the Guardian on 12 February 2026, treatment resistance was not the reason given. The reason given was continuing substance use: the service had to "plan around the realities he's likely to continue using," and needed him "to stay well even if he uses."

That prophylactic rationale cannot survive the rest of the same conversation. In it, the consultant stated that clearance of olanzapine and clozapine is accelerated by smoking and "more pronounced" in this patient than in general, and that "the reports say that he's likely to be a nonresponder to these medications like olanzapine and clozapine because you can't reach a very good level in the blood." A drug that cannot reach a therapeutic concentration cannot keep anyone well while they use. The patient absconds and reduces his own medication, so the exposure is interrupted as a matter of established pattern. The Guardian served a Cease and Desist in that same conversation, and it was acknowledged. **No one with authority to consent, consented.**

What the record documents is therefore not resistance but its manufacture. Each escalation produced the supersensitivity that made the next relapse worse; each relapse was read as confirmation of the label;

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[2] Howes OD, McCutcheon R, Agid O, de Bartolomeis A, van Beveren NJM, Birnbaum ML, et al. *Treatment-Resistant Schizophrenia: Treatment Response and Resistance in Psychosis (TRRIP) Working Group Consensus Guidelines on Diagnosis and Terminology*. Am J Psychiatry. 2017;174(3):216–229. doi:10.1176/appi.ajp.2016.16050503

[3] Cipolla et al., *Combination of Two Long-Acting Antipsychotics in Schizophrenia Spectrum Disorders: A Systematic Review*, *Brain Sciences*, 2024

the label authorised the next escalation. The harm is iatrogenic at every step, and the label is the instrument by which it is recorded as illness.

B. The pharmacogenomic mechanism of manufactured resistance

The mechanism is not speculative; it is measurable, and in the exemplar case it was measured — by the treating team, on 12 February 2026:

- On the treating team's own genetic panel, reported to the Guardian by the treating consultant on 12 February 2026, the patient is a **metaboliser whose enzymes are prone to induction**, so that smoking accelerates clearance of **olanzapine and clozapine** — and a **predictable metaboliser of aripiprazole**, for whom blood levels can be forecast. In the consultant's own words, smoking "can lower olanzapine and clozapine levels, but in his case, it will be more pronounced".
- Metabolism of that class is further **induced by tobacco and cannabis**, principally via CYP1A2. Heavy smoking reduces plasma concentrations of clozapine by ~50% and olanzapine by ~67%; smoking cessation can triple plasma clozapine (Haslemo et al., *Eur J Clin Pharmacol* 2006; Bondolfi et al., *Ther Drug Monit* 2005)[4].
- This is not a rare idiosyncrasy. In a 2025 cohort (n=453), **89.6%** carried the inducible CYP1A2 allele and **62.9%** carried a CYP2D6/CYP3A4 allele altering metabolism (Varney et al., *Pharmaceuticals* 2025;18(6):892)[5]. Metaboliser status is common, testable, and — in the exemplar — was ignored.

The closed loop follows mechanically: rapid clearance → sub-therapeutic level at standard dose → apparent non-response → dose escalated toward maximum → iatrogenic toxicity **without** therapeutic benefit → continued smoking sustains the induction → the record reads "treatment resistant" → clozapine. Every turn of that loop is legible as a pharmacokinetic event, not a psychiatric one.

The treating consultant drew that conclusion himself, in terms, before the drug was given. Reporting the panel to the Guardian on 12 February 2026, he said that "the reports say that he's likely to be a nonresponder to these medications like olanzapine and clozapine because you can't reach a very good level in the blood." Clozapine was commenced regardless, despite Cease and Desist notice given by the Guardian and patient, escalated to 550 mg, and no plasma level has been produced. **The finding this submission asks a Royal Commission to investigate was made by the prescriber, recorded contemporaneously, and then disregarded by the person who made it**

C. Dopamine Receptor Supersensitivity (DRS) — the engine of relapse

When D2 receptors are chronically blocked by tight antagonists (zuclopenthixol, haloperidol, olanzapine), the brain compensates by up-regulating D2 receptor **density and sensitivity**. When the blockade falls — drug stopped, dose reduced, depot washing out, plasma level dropping with a change in smoking, or absconding — an excess of hypersensitive receptors meets normal dopamine, and the result is a rebound

[4] Haslemo et al., *Eur J Clin Pharmacol* 2006; Bondolfi et al., *Ther Drug Monit* 2005
[5] Varney et al., *Pharmaceuticals* 2025;18(6):892

psychosis that can exceed the original episode (Chouinard et al., *Antipsychotic-induced dopamine supersensitivity psychosis: pharmacology, criteria, and therapy*, *Psychotherapy and Psychosomatics* 2017;86(4):189–219)[6].

Supersensitivity is not fringe. Reported prevalence is on the order of **30% in schizophrenia and ~70% in "treatment-resistant" schizophrenia** (Chouinard et al., PPS 2017) — which raises the possibility that a large fraction of what is labelled TRS *is* iatrogenic DRS: a condition created by the treatment, then cited as proof the illness is intractable. A fall in plasma concentration produces a fall in D2 blockade indistinguishable in effect from abrupt dose reduction, which is causally associated with relapse via supersensitivity persisting months to years (Horowitz, Jauhar, Natesan, Murray & Taylor, *Schizophrenia Bulletin* 2021;47(4):1116–29)[7].

Nor is the concept confined to overseas research literature. In 2022 the journal of the Royal Australian and New Zealand College of Psychiatrists published a reappraisal arguing that chronic antipsychotic exposure induces neuroadaptive changes in dopaminergic and other systems which render some individuals more vulnerable to psychotic relapse — **including those receiving continuous antipsychotic treatment** — and that in treating every episode of psychosis with prolonged or indefinite therapy, the profession may paradoxically increase the risk of relapse in a significant proportion of people. Its author concluded that it was time greater attention was paid to supersensitivity and that practice be adjusted accordingly (Lugg, Aust N Z J Psychiatry 2022;56(5):437–444).

The clinical implication is decisive: for supersensitivity, **more blockade is the wrong answer**. The indicated course is a partial agonist that occupies D2 without shutting it down (aripiprazole), permitting gradual receptor normalisation, combined with a slow, hyperbolic taper — never rapid escalation of a new full antagonist on an acute ward.

This is not merely a preference between agents. Aripiprazole appears not to build the trap in the first place: chronic aripiprazole treatment **prevents the development of dopamine supersensitivity** and, potentially, supersensitivity psychosis (Tadokoro et al., *Schizophrenia Bulletin* 2012;38(5):1012–1020)[8], consistent with its very low rates of tardive dyskinesia in comparative randomised trials (Carbon et al., *World Psychiatry* 2018;17(3):330–340)[9]. The pharmacogenomically indicated agent in the exemplar case is therefore also the agent least likely to manufacture the resistance the system then names.

D. Genotype is not phenotype — the variable nobody measured

CYP1A2 is **constitutively expressed**. It is always present and always metabolising clozapine. The gene does not switch metabolism on or off; it governs **how far enzyme production rises in response to exposure**.

The inducing agents are **polycyclic aromatic hydrocarbons — the combustion products of tobacco and smoked cannabis** — acting through aryl hydrocarbon receptor-mediated transcriptional upregulation. Neither nicotine nor THC is the inducer; combustion is. This has now been demonstrated directly in human liver tissue: cigarette smoke extract produces reliable, concentration-dependent induction

[6] Chouinard et al., *Antipsychotic-induced dopamine supersensitivity psychosis: pharmacology, criteria, and therapy*, *Psychotherapy and Psychosomatics* 2017;86(4):189–219
[7] Horowitz, Jauhar, Natesan, Murray & Taylor, *Schizophrenia Bulletin* 2021;47(4):1116–29
[8] Tadokoro et al., *Schizophrenia Bulletin* 2012;38(5):1012–1020
[9] Carbon et al., *World Psychiatry* 2018;17(3):330–340

of CYP1A1 and CYP1A2 across multiple primary human hepatocyte donors, with CYP1A1 enzyme activity induced at least 57-fold (Lenich & Ruez, *Archives of Toxicology* 2025;99:4513–4530)[10].

The CYP1A2*1F allele (rs762551, −163C>A) confers high inducibility rather than elevated baseline activity. Sachse et al. found no significant genotype effect among non-smokers but a 1.6-fold activity increase in A/A smokers, and suggested the A allele might be recessive for this trait — a question their 51 smokers were underpowered to settle *(Sachse et al., Br J Clin Pharmacol 1999)*[11]. The larger meta-analysis resolves it. Across 3,570 subjects, heterozygotes showed significantly higher CYP1A2 activity than wild-type (SMD 0.32, 95%CI 0.11–0.54, p=0.003), rising among smokers to SMD 0.56 (95%CI 0.22–0.90, p=0.001). No other CYP1A2 polymorphism showed any effect (Koonrungsesomboon et al., The Pharmacogenomics Journal 2018;18(6):760–768)[12]. Phenotype is therefore the product of genotype and exposure — demonstrable in heterozygotes, but only where an inducer is present.

Consistency with the treating team's own finding. Raw genotype data (AncestryDNA V2.0, build 37, forward strand, 3 November 2023) shows rs762551 = A/C — heterozygous carriage of CYP1A2*1F. Whether a single copy is sufficient remains unsettled on the evidence set out above. The array therefore corroborates, at the level of genotype, what the treating team recorded at the level of phenotype — baseline metabolism, prone to induction — and it does not need to do more than that. **The induction itself does not depend on genotype at all: CYP1A2 is constitutively expressed and inducible in anyone who combusts plant material, as the hepatocyte data above establish across multiple donors with no genotype selection. The genotype governs how far, not whether**. The genotype is consistent with the treating team's own recorded finding that his metabolism is prone to induction (the treating consultant, 12 February 2026). That clinical pharmacogenomic report, not the consumer array, remains the primary evidence.

The consequence for dosing. At a **fixed prescribed dose**, plasma concentration is a moving target — suppressed during unrestricted smoking, rising when access is restricted. Milligrams prescribed convey no reliable information about drug exposure. In a patient whose smoking has varied continuously across open wards, leave, absconding and HDU placement, **dose and exposure are effectively uncoupled**.

The deinduction hazard — a safety failure, not merely an efficacy one. Induction reverses when exposure ceases. Enzyme activity begins to fall within 24 hours; a new, lower steady state is typically reached at **one week**; de-induction is generally complete at two to four weeks. The apparent half-life of the decline is approximately **38.6 hours** (Faber MS, Fuhr U, Clin Pharmacol Ther 2004;76(2):178–184)[13]. Smoking cessation can produce a **threefold rise in plasma clozapine on an unchanged dose** (Bondolfi et al., *Ther Drug Monit* 2005)[14]. A patient escalated toward maximum dose while smoking exposure falls is on a pharmacokinetic collision course. Clozapine toxicity following smoking cessation is recognised, documented, and foreseeable.

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[10] Lenich & Ruez, *Archives of Toxicology* 2025;99:4513–4530

[11] Sachse C, Brockmöller J, Bauer S, Roots I. *Functional significance of a C→A polymorphism in intron 1 of the cytochrome P450 CYP1A2 gene tested with caffeine.* Br J Clin Pharmacol. 1999 Apr;47(4):445–449. doi: 10.1046/j.1365-2125.1999.00898.x · PMID 10233211

[12] Koonrungsesomboon N, Khatsri R, Wongchompoo P, Teekachunhatean S. *The impact of genetic polymorphisms on CYP1A2 activity in humans: a systematic review and meta-analysis.* The Pharmacogenomics Journal 2018;18(6):760–768. Published online 27 December 2017.

[13] Faber MS, Fuhr U. *Time response of cytochrome P450 1A2 activity on cessation of heavy smoking.* Clin Pharmacol Ther 2004;76(2):178–184)

[14] Bondolfi et al., *Ther Drug Monit* 2005

But the exposure variable is not merely unmeasured — it is actively mis-assumed. Wards treat admission as a smoking-cessation event and record patients accordingly. Two things happen instead. Where leave permits, patients simply keep smoking: in a prospective study of 31 clozapine-treated inpatients through a hospital-grounds smoking ban, neither reported tobacco consumption nor clozapine and norclozapine plasma concentrations changed, because most patients held off-grounds leave (Gee SH, Taylor DM, Shergill SS, Flanagan R, MacCabe JH, *Ther Adv Psychopharmacol* 2017;7(2):79–83)[15]. Where leave does not permit it, they substitute.

In Australian custodial settings the practice is sufficiently established to have acquired a name — "teabacco", tea leaves rolled as a tobacco substitute following institutional bans (Puljević C et al., *Drug and Alcohol Review* 2018)[16]. In the exemplar case the Guardian observed the patient smoking black tea leaves rolled in paper at Logan Hospital in July 2026, and he has confirmed continuing the practice on the current admission.

The inducing agent is the polycyclic aromatic hydrocarbon content of combusted plant material — not nicotine, and not the species. Nicotine replacement, chewing tobacco and e-cigarettes do not produce the interaction; combustion does (Mould & Hogan, Oxford Health NHS Foundation Trust, December 2023; Haslemo T et al., *Eur J Clin Pharmacol* 2006;62(12):1049–1053). Herbal (tobacco-free) cigarettes yield PAHs at levels comparable to or exceeding tobacco, with benzo(α)pyrene measured higher in an herbal (tobacco-free) cigarette than in a standard cigarette of **equal tar** (Bak JH et al., *Toxicological Research* 2015;31(1):41–48; Gilbert & Lindsey, *Br J Cancer* 1956;10(4):646–648)[17] [18]. Campbell and Lindsey measured PAH in unburnt material and found far less than in smoke — which establishes that PAHs are formed by combustion, not merely released from the plant. That is the mechanistic keystone of the argument: if burning creates the PAHs, the species of leaf is largely beside the point (Campbell & Lindsey, Br J Cancer 1956;10(4):649–652)[19]. Cannabis demonstrates the principle conclusively: THC is a CYP2C9/CYP3A4 substrate and not a CYP1A2 substrate, yet cannabis smoke induces CYP1A2 and cannabis cessation raises clozapine plasma levels (Zullino et al., Int Clin Psychopharmacol 2002;17(3):141–143)[20]. Induction is proportional to PAH bioavailability and is greater with unfiltered smoking; as few as 7–12 cigarettes daily may achieve maximal induction (Mould & Hogan, Oxford Health NHS Foundation Trust, December 2023; Haslemo T et al., *Eur J Clin Pharmacol* 2006;62(12):1049–1053)[21] [22].

In the exemplar case the patient smoked unfiltered throughout, on the Guardian's direct and contemporaneous observation, and no enquiry as to cigarette type appears anywhere in the record.

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[15] Gee SH, Taylor DM, Shergill SS, Flanagan R, MacCabe JH, Ther Adv *Effects of a smoking ban on clozapine plasma concentrations in a nonsecure psychiatric unit.* PMID 28255437. Psychopharmacol 2017;7(2):79–83)

[16] Puljević C, et al. *'Teabacco': smoking of nicotine-infused tea as an unintended consequence of prison smoking bans. Drug and Alcohol Review.* 2018;37(7). doi:10.1111/dar.12848

[17] Bak JH, Lee SM, Lim HB. *Safety assessment of mainstream smoke of the herbal cigarette.* Toxicological Research. 2015;31(1):41–48. doi:10.5487/TR.2015.31.1.041 · PMID 25874032

[18] Gilbert JAS, Lindsey AJ. *Polycyclic hydrocarbons in tobacco smoke: pipe smoking experiments.* Br J Cancer. 1956 Dec;10(4):646–648. doi:10.1038/bjc.1956.77 · PMID 13426375

[19] Campbell JM, Lindsey AJ. *Polycyclic hydrocarbons extracted from tobacco: the effect upon total quantities found in smoking.* Br J Cancer. 1956 Dec;10(4):649–652. doi:10.1038/bjc.1956.78 · PMID 13426376

[20] Zullino DF, Delessert D, Eap CB, Preisig M, Baumann P. *Tobacco and cannabis smoking cessation can lead to intoxication with clozapine or olanzapine.* Int Clin Psychopharmacol. 2002 May;17(3):141–143. doi:10.1097/00004850-200205000-00008 · PMID 11981356

[21] Mould S, Hogan R. *Guidance on the effects of smoking and smoking cessation on psychotropic and other medications.* Oxford Health NHS Foundation Trust, Drugs & Therapeutics Group; approved December 2023.

[22] Haslemo T, Eikeseth PH, Tanum L, Molden E, Refsum H. *The effect of variable cigarette consumption on the interaction with clozapine and olanzapine.* Eur J Clin Pharmacol. 2006;62(12):1049–1053. doi:10.1007/s00228-006-0209-9

The magnitude of induction from smoked tea specifically has not been quantified in the published literature. Every determinant of induction, however, is present: combustion of plant material, unfiltered delivery, daily use, and a low saturation threshold. His clearance has therefore probably remained induced throughout the present titration, whatever his chart records.

The consequence is documented, not theoretical. In a retrospective evaluation of a total smoking ban on a psychiatric inpatient unit, 4.2% of smoking patients had plasma clozapine concentrations of 1000 μg/L or above before the ban; within six months of it, 41.7% did, despite dose reductions (Cormac I, Brown A, Creasey S, Ferriter M, Huckstep B, Acta Psychiatr Scand 2010;121(5):393–397). A cohort, not an anecdote: an institutional smoking ban pushed four in ten patients into the toxic range. Where plasma levels are not measured, such episodes are recorded as illness rather than as pharmacokinetics.[23]

A second, opposing variable runs simultaneously. Caffeine is metabolised almost entirely by CYP1A2 and competes with clozapine for it, tending to **raise** plasma levels. Institutional settings are characterised by heavy coffee consumption. The patient therefore sits between two large, opposing, wholly unmonitored influences on his plasma concentration — one suppressing it, one elevating it — while his dose is escalated on the assumption that milligrams determine exposure.

This is the core systemic finding. A service that does not closely measure plasma concentration, does not verify smoking status, and does not account for dietary CYP1A2 competition is not titrating to effect. It is titrating blind, and recording the result as a property of the patient.

What resolves all of it: therapeutic drug monitoring. Every hazard above is invisible without plasma clozapine and norclozapine levels, and every one is resolvable with them. The **clozapine:norclozapine ratio** additionally distinguishes rapid metabolism from non-adherence — a distinction of direct forensic significance where absconding is on the record, and one that cannot be drawn from dose or behaviour. TDM is not routine practice in Australian public psychiatry.

None of this is esoteric, and none of it is new. The Government of South Australia's Department for Health and Wellbeing publishes two pages of clinical guidance titled *Factors influencing clozapine plasma concentrations* (last updated March 2023). It states that therapeutic drug monitoring is indicated where there is **poor compliance, inadequate response to treatment, significant adverse effects, or possible drug interactions**. The exemplar patient met all four criteria, simultaneously and continuously, for the whole of his exposure. It states that tobacco smoke induces CYP1A2 and lowers clozapine concentrations by up to 50%; that nicotine replacement does **not** induce the enzyme, the interaction being "mediated by inhaling the smoke itself"; that **seven to twelve cigarettes daily** is probably sufficient to induce clozapine metabolism; that caffeine inhibits CYP1A2 and may raise clozapine concentrations, with care required for patients consuming coffee, tea or cola; and that the usual target range is **350–600 μg/L**.

It also contains this instruction:

> *"Care must be taken to monitor patients closely after they cease smoking during clozapine treatment, as plasma levels may increase by up to 50%, leading to increased adverse effects, notably seizures."*

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[23] Cormac I, Brown A, Creasey S, Ferriter M, Huckstep B. *A retrospective evaluation of the impact of total smoking cessation on psychiatric inpatients taking clozapine.* Acta Psychiatr Scand. 2010 May;121(5):393–397. doi:10.1111/j.1600-0447.2009.01482.x · PMID 19824991

Four independent predictors of low plasma concentration were present, and none was measured. The same two pages identify the factors that lower clozapine plasma concentrations. In the exemplar case four were present simultaneously and continuously:

- **Male sex** — "males seem to metabolise clozapine faster than females and achieve lower average plasma concentrations";
- **Age** — patients aged 45–54 have been found to have *higher* plasma levels than those aged 18–26, attributed to reduced liver enzyme activity in the older group. He was eighteen when olanzapine depot was prescribed;
- **Smoking** — up to 50% lower, at seven to twelve cigarettes daily;
- **Inducible genotype** — CYP1A2*1F carriage, present in his own 2023 raw genotype data and corroborated by the treating team's own recorded finding of induction-proneness in February 2026.

Each of these pushes exposure in the same direction: down. Their combined magnitude in an individual patient cannot be derived from first principles — which is precisely why the guidance directs that it be measured. It was not measured. The dose was escalated instead, and the resulting non-response was recorded as a property of the patient rather than of his pharmacokinetics.

The guidance also closes the remaining escape route. It notes that some patients "may experience limiting adverse effects despite concentrations in the recommended range." A level within 350–600 μg/L would therefore not have excused the seizure, the cardiac presentation, or the reported pleural effusion[24]. But no plasma-level record was ever produced — so the service cannot say the concentration was appropriate either. It does not know. It chose not to know.

An Australian health department told prescribers, in plain language and in two pages, that smoking cessation during clozapine treatment can raise plasma levels by half and cause seizures. In the exemplar case a seizure occurred at 400 mg. No plasma level has been produced in answer to his requests, or to the Guardian's. The failure documented in this submission is not a failure to know something difficult. It is a failure to apply guidance written for exactly this situation, published by a health department of the same country, and freely available (Government of South Australia, Department for Health and Wellbeing, March 2023)[25].

The same protocol exists in a second health system, and it addresses his precise circumstance. Oxford Health NHS Foundation Trust guidance, approved December 2023, deals directly with unplanned smoking cessation on admission to an in-patient unit. It directs clinicians to determine smoking status and concordance, to review previous plasma levels, to consider a 30% dose reduction where a patient ceases smoking on an unchanged dose, to check a plasma level within a week — and, decisively here, that **where a patient is re-titrated on clozapine the new dose target may need to be 30% lower than the previous dose, with plasma levels monitored as the dose increases** (Mould & Hogan, Oxford Health NHS Foundation Trust, December 2023)[26]**.**

The exemplar patient was re-titrated on clozapine from 9 July 2026 following readmission. His previous course had reached 550 mg. The guidance for that precise situation directs a target approximately 30%

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[24] Mouaffak F, Gaillard R, Burgess E, Zaki H, Olié JP, Krebs M-O. *Clozapine-induced serositis: review of its clinical features, pathophysiology and management strategies*. Clin Neuropharmacol 2009;32(4):219–223. doi:10.1097/wnf.0b013e318197a2f2 · PMID 19620851 — polyserositis; discontinue immediately and do not rechallenge. [F]
[25] Government of South Australia, Department for Health and Wellbeing. *Factors influencing clozapine plasma concentrations*. Clinical guidance, last updated March 2023 (www.sahealth.sa.gov.au)
[26] Mould S, Hogan R. *Guidance on the effects of smoking and smoking cessation on psychotropic and other medications*. Oxford Health NHS Foundation Trust, Drugs & Therapeutics Group; approved December 2023.

lower, with monitoring throughout. He was escalated toward approximately 500 mg in twelve days, and no plasma level is known to have been taken at any stage.

The same guidance identifies enzyme saturation as a distinct hazard and names the patients in whom it should be suspected. It warns that CYP1A2 can become saturated, after which the rate of clozapine metabolism falls dramatically and small increases in dose produce vast increases in concentration. It directs that this be considered in patients with high serum concentrations, or those particularly susceptible to dose-related adverse effects — **giving a history of seizures as its example.** The exemplar patient had a seizure on 400 mg on 18 March 2026. He was re-titrated toward 500 mg four months later without a plasma level. The guidance separately warns that high doses themselves increase saturation risk, producing large and unpredictable rises.

It also requires the very datum that was never collected. The first of its general recommendations is to ascertain baseline smoking status, quantity, and **type of cigarette — specifically whether filtered or unfiltered** — because induction is proportional to PAH bioavailability and unfiltered smoking produces more of it. The exemplar patient was smoking unfiltered hand-rolled tea. No such enquiry appears anywhere in the record.

A third statement of the same protocol comes from the pharmacy department that writes the Maudsley Prescribing Guidelines. The authors of the smoking-ban study set out above close it with recommendations for practice: baseline plasma concentrations **before**, and **weekly after**, any change in smoking habit, with doses adjusted until a concentration in the therapeutic range is reached; and consideration of predicted smoking status **after discharge**, because concentrations should be expected to fall again when smoking resumes (Gee et al., *Ther Adv Psychopharmacol* 2017) [27].. Three health systems — South Australia, Oxford, and the Maudsley — describe the same monitoring, in the same terms, for the same drug. The exemplar patient's smoking status changed on every admission, every period of leave, every absconding and every readmission across thirty-four months. No plasma level has been produced for any of them.

In the exemplar case, the Guardian has been unable to establish that **a single plasma clozapine level was ever taken** across a course reaching 550 mg, because the record has been withheld for eleven months across five statutory requests.

The fluctuation recorded as a deteriorating, treatment-resistant illness has a measurable pharmacokinetic cause that was never measured. That is the generalisable finding: wherever therapeutic drug monitoring is absent, pseudo-resistance is indistinguishable from true resistance — and the label defaults to the patient.

E. What the label concealed was a treatable case

Every argument above is about what the treatment did. This one is about what it displaced.

The exemplar patient's recorded primary condition is substance use disorder. Beneath it sit complex post-traumatic stress and an untreated childhood trauma. Neither is contested by the service: its own discharge summary of 15 July 2025 records "poor response to multiple antipsychotic medications" and, in the same document, recommends drug-and-alcohol treatment

[27] Gee SH, Taylor DM, Shergill SS, Flanagan R, MacCabe JH, Ther Adv *Effects of a smoking ban on clozapine plasma concentrations in a nonsecure psychiatric unit.* PMID 28255437. Psychopharmacol 2017;7(2):79–83)

and rehabilitation. **The service named the conditions and named the treatments. Across thirty-four months and sixteen admissions, it provided neither**. What it provided was seven antipsychotic agents, lithium, benzodiazepines and an anticholinergic, in escalating combination.

A dopamine antagonist does not treat trauma. It cannot. It has no mechanism by which to do so, and no guideline proposes one. What it can do is suppress the expression of distress — flatten affect, slow speech, sedate, immobilise — so that the presentation becomes quieter while the condition producing it remains exactly where it was, untouched, and accumulating. That is not treatment. It is the management of a signal.

The consequence is that the harm compounds in both directions at once. The unaddressed trauma continues to generate the presentation, and each escalation is read as a response to the presentation rather than to its cause — so the record accumulates evidence of an intractable illness while the actual condition is never once the subject of a treatment plan. Meanwhile the drugs generate their own pathology: supersensitivity, parkinsonism, metabolic collapse, cardiac events, and a young man who cannot rise before midday and cannot look at photographs of himself. **Some of that is now irreversible. Drug-induced parkinsonism persists or progresses in 10–50% of cases after the drug is withdrawn (Shin & Chung, J Clin Neurol 2012). Tardive dyskinesia may not remit at all. Three years of prefrontal maturation cannot be re-run.**

This is the finding the label conceals, and it is why the failure to reach a differential diagnosis is not a procedural defect but the injury itself. A differential is not paperwork. It is the point at which a clinician decides what they are treating — and everything downstream, every dose, every escalation, every year of exposure, follows from it. Where that decision is never made, the treatment cannot be aimed. It can only be increased. **The exemplar patient was not treated for the wrong condition. He was treated for no condition at all, and the increase was mistaken for the care.**

PART II. OVERMEDICATION AND RAPID LOADING

The escalation-rate problem

Standard clozapine initiation is deliberately slow: 12.5 mg once or twice on day one, titrated cautiously, with therapeutic drug monitoring, toward a target of roughly 300–450 mg — and ancestry- and smoking-adjusted where indicated (US package insert; *Psychiatric News* clozapine update, 2025; StatPearls)[28]. The slowness is not caution for its own sake; it is dictated by the risk profile — myocarditis characteristically emerging days 10–33 (Ronaldson et al., *Aust NZ J Psychiatry* 2011)[29] with 83% developing between days 14–21, seizure threshold falling in a dose-related manner, and cardiovascular and inflammatory reactions clustering in the first weeks.

Against that baseline, the exemplar case shows loading measured in **days**, not weeks:

- Escalation, current admission (contemporaneous carer observation, pending records): from 15 mg on 9 July 2026 rising to approximately 500–550 mg by 21 July 2026 — a span of 12 days. That is an increase of the order of thirty-fold in twelve days, with the largest proportional increases occurring in the first few steps. *(Exact daily increments are the Guardian's contemporaneous observation; the complete eMeds record has been withheld for eleven months and is required to fix the precise curve — the direction and speed are not in doubt.)*

- **Prior course (documented):** the February–March 2026 clozapine course reached **550 mg**, produced a **seizure** (18 March 2026, on 400 mg with lorazepam and aripiprazole), and included a **double-dose administration error — 800 mg instead of 400 mg** (22 March 2026), which the patient vomited.

The comparison a Royal Commission should weigh

Safe **reduction** of antipsychotics is measured at **2–10% per fortnight, hyperbolically, over months** (Horowitz et al., 2021)[30]. The exemplar shows loading proceeding at a rate that, expressed per fortnight, exceeds the safe reduction rate by orders of magnitude — a comparison the Commission can make directly from the two figures. This is the definition of overmedication: **maximal exposure to toxicity, minimal prospect of benefit**.

The diagnosis changes the titration target

A further failure sits beneath the rate. The optimal D2 receptor occupancy window that underpins standard antipsychotic dosing was derived from patients with **normal receptor density**. In dopamine supersensitivity that assumption fails: density is up-regulated by the prior blockade, so an identical percentage occupancy leaves a materially larger absolute number of receptors available to endogenous

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[28] Haidary HA, Padhy RK. *Clozapine.* In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; last updated 10 November 2023. Also: Special Report: *Clozapine Revisited.* Psychiatric News. American Psychiatric Association; 2025.

[29] Ronaldson KJ, Fitzgerald PB, Taylor AJ, Topliss DJ, McNeil JJ. *A new monitoring protocol for clozapine-induced myocarditis based on an analysis of 75 cases and 94 controls.* Aust N Z J Psychiatry 2011;45(6):458–465 — onset 10–33 days, 83% between days 14 and 21; nine deaths; protocol recommends baseline troponin I/T, CRP and echocardiography, then troponin and CRP on days 7, 14, 21 and 28. [F]

[30] Horowitz MA, Jauhar S, Natesan S, Murray RM, Taylor D. *A method for tapering antipsychotic treatment that may minimize the risk of relapse.* Schizophrenia Bulletin. 2021;47(4):1116–1129.

dopamine. Iyo et al. accordingly identify the conventional occupancy index as of limited application in DSP, and propose that adequate neurotransmission is governed by the **number of receptors available for binding** rather than by the proportion blocked (Iyo et al., J Clin Psychopharmacol 2013;33(3):398–404)[31].

The management principle that follows is stability, not escalation. In the largest prospective series of its kind — 108 treatment-resistant patients, 72 of them with a documented DSP history — the therapeutic strategy was to maintain thoroughly stable D2 occupancy, with oral agents switched **partially and gradually** to long-acting injectable (Kimura et al., J Psychopharmacol 2016)[32]. Rapid escalation of a full antagonist is the opposite of that strategy on both limbs: it maximises fluctuation, and it increases blockade.

The second diagnostic criterion for supersensitivity is the very finding relied upon to justify escalation. DSP is defined by rebound psychosis following the reduction, withdrawal or switching of an antipsychotic, **and/or by acquired tolerance — the antipsychotics ceasing to control symptoms even at an increased dose**. A patient who fails to respond to successive dose increases therefore satisfies a diagnostic criterion for supersensitivity. In the exemplar case that failure was recorded instead as evidence of treatment resistance, and answered with further escalation.

The distinction is not academic. On one reading the dose should rise; on the other it should stabilise or fall. **A differential diagnosis that was never made determined the direction of every dose decision that followed.**

The limit of the claim is stated deliberately. No published guideline specifies a numerical titration rate for supersensitivity psychosis, and none is asserted here. What the literature establishes is narrower and sufficient: occupancy targets derived from normal-density patients do not transfer; the therapeutic goal is stable rather than higher occupancy; and switching should be partial and gradual. Each of those three propositions was contradicted by the course actually taken.

The formulation was wrong before the dose was. Clozapine is an oral medication requiring uninterrupted daily administration; where more than forty-eight hours of doses are missed it must be re-titrated from the beginning. At the time clozapine was commenced, the record documented sixteen admissions, including absconding and being re-admitted, and a consistent pattern of self-deprescribing. He has absconded three times in the present admission alone, since clozapine was recommenced. The service prescribed, to a documented serial absconder, the one drug whose interruption is most dangerous — and it demonstrated that it knew the consequence, by re-titrating him from 15 mg on 9 July 2026. Each such cycle reopens the myocarditis window of days 10 to 33, with 83% of cases arising between days 14 and 21 (Ronaldson et al.), and each therefore triggers the weekly troponin and C-reactive protein monitoring mandated by NSW Health Guideline GL2022_011. No such series has been produced for any cycle. The choice of formulation was not neutral. It converted a known and documented behavioural pattern into recurrent cardiac risk.

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[31] Iyo M, Tadokoro S, Kanahara N, Hashimoto T, Niitsu T, Watanabe H, Hashimoto K. *Optimal extent of dopamine D2 receptor occupancy by antipsychotics for treatment of dopamine supersensitivity psychosis and late-onset psychosis*. J Clin Psychopharmacol. 2013;33(3):398–404. doi:10.1097/JCP.0b013e31828ea95c · PMID 23609386.
[32] Kimura H, Kanahara N, Sasaki T, et al. *Risperidone long-acting injectable in the treatment of treatment-resistant schizophrenia with dopamine supersensitivity psychosis: results of a 2-year prospective study, including an additional 1-year follow-up*. J Psychopharmacol. 2016. doi:10.1177/0269881116655978.

Poly-pharmacy and the count of agents

Since September 2023, the exemplar patient has been exposed to at least **seven antipsychotic agents** — aripiprazole, brexpiprazole, paliperidone, haloperidol, olanzapine, zuclopenthixol, clozapine (two of them first-generation) — plus lithium, benzodiazepines (diazepam, lorazepam) and benztropine, in combinations and abrupt substitutions. He was **seventeen** at first admission and seventeen when the first depot — paliperidone — was given, on 19 October 2023, one month before his eighteenth birthday. He was discharged from his **first drug-induced presentation with five benzodiazepine tablets** handed to him and no taper. The pharmacogenomically indicated agent — aripiprazole — has never been adopted as the primary drug, and on the current admission has reportedly been **ceased.**

One claim commonly made about poly-pharmacy is not advanced here, and the reason matters more than the claim. Long-term antipsychotic poly-pharmacy is widely assumed to increase mortality relative to monotherapy. The meta-analysis does not support that: across 12 studies and 834,534 person-years there was no difference in all-cause mortality between poly-pharmacy and monotherapy (crude rate ratio 0.94, 95% CI 0.81–1.10, p=0.446; adjusted HR 0.98, 95% CI 0.80–1.19, p=0.802), and the authors record that practice guidelines nonetheless advise against long-term poly-pharmacy on efficacy rather than mortality grounds (Buhagiar et al., *Schizophr Res* 2020)[33].

But that comparison is drawn entirely inside the exposed population. It compares one arm of treatment with another. It contains no unexposed arm, and it therefore cannot say whether either arm is safe — only that they resemble each other. What sits outside the comparison is the figure that does not move: a life expectancy gap of 15–20 years for schizophrenia (Correll et al. 2022) and 20.38 years for substance use disorder (Chan et al. 2023), unchanged across thirty years and across the entire era of second-generation drugs (Tanskanen et al. 2018). One drug or five, the gap is the same size. A null result between two arms of the same exposure is not a finding of safety. It is a finding that the question was not asked.

Nor can the registers answer it. Roughly four in five deaths in this population are recorded as natural-cause, led by cardiovascular disease (Yeh et al. 2023: 6,176 of 7,730 deaths, 79.9%). Death registers code the organ that failed, not the exposure that preceded it. Any treatment-related contribution to that gap is therefore distributed across cause-of-death categories that no register attributes to treatment, and no coroner is asked to. Correll and colleagues nonetheless list antipsychotic use among the **modifiable** contributors to the gap — which is to say that the literature contemplates a treatment-related component and possesses no instrument capable of sizing it.

The objection advanced here is therefore the narrow one, and it is sufficient. Guidelines advise against long-term poly-pharmacy; efficacy in combination is unestablished; each additional agent adds adverse-effect burden; and each additional agent added to manage the effects of the last compounds the burden it was added to relieve. What proportion of a twenty-year gap is attributable to that architecture is a question no one has asked of any Australian dataset. **It is Term of Reference 6, *Outcomes audit*, below.**

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[33] Buhagiar K, Templeton G, Blyth H, Dey M, Giacco D. *Mortality risk from long-term treatment with antipsychotic polypharmacy versus monotherapy among adults with serious mental illness: a systematic review and meta-analysis of observational studies.* Schizophr Res 2020. doi:10.1016/j.schres.2020.08.026

PART III. RAPID WITHDRAWAL AND THE ABSENCE OF TAPERING

The evidence on how to stop

The single most consequential figure in this submission is the relapse gradient by withdrawal method. A systematic review and meta-analysis found an inverse dose-response between the duration of dose reduction and psychotic relapse: abrupt cessation, event rate **0.77** per person-year (95% CI 0.56–0.98); reduction over 1–2 weeks, **0.57** (0.35–0.80); over 3–10 weeks, **0.47** (0.28–0.67); over more than ten weeks, **0.31** (0.26–0.36); P < .0001. (Bogers et al., *Schizophrenia Bulletin Open* 2020;1(1):sgaa002, as reported in Horowitz et al., *Schizophrenia Bulletin* 2021)[34].

Duration of dose reduction	Cohorts	Relapse event rate per person-year (95% CI)
Abrupt (0 weeks)	14	0.77 (0.56–0.98)
1–2 weeks	12	0.57 (0.35–0.80)
3–10 weeks	7	0.47 (0.28–0.67)
More than 10 weeks	10	0.31 (0.26–0.36)

Bogers et al., Schizophr Bull Open 2020;1(1):sgaa002. Duration strata cover 43 of the 46 cohorts; three cohorts did not report duration of dose reduction. P < .0001 across strata.

Two features of the same analysis should be stated rather than left to be found. The protection lies in reducing to a maintained dose rather than in stopping slowly: gradual discontinuation carried an event rate of 0.92 (0.58–1.26), higher than abrupt discontinuation at 0.77, while gradual reduction to a maintained rest dose carried 0.35 (0.25–0.44). And the authors' own conclusion is that discontinuation is not advised in chronic schizophrenia, and that reduction should be gradual and preferably not below 5 mg haloperidol equivalent. Neither point weakens the case here. The exemplar was never reduced at all — not to a rest dose, not slowly, not once across sixteen admissions — and the method actually used, abrupt cessation, sits at the adverse end of every stratum in the table.

Two corollaries follow, and both indict current practice. First, **a four-week taper is not a taper**: meta-analysis found no significant benefit of "gradual" discontinuation over four weeks compared with stopping abruptly (Leucht et al., *Lancet* 2012)[35], whereas tapering over three to nine months **halved** the relapse rate (Viguera et al., *Arch Gen Psychiatry* 1997)[36]. Second, relapse is not distributed evenly after cessation but clusters around it: **60% of all relapses over four years occurred within three months of stopping** (Viguera 1997, as reported in Horowitz, Murray & Taylor, *JAMA Psychiatry* 2020). A disease that relapses at random does not cluster. A withdrawal state does. Safe reduction must be **hyperbolic** — proportionally smaller cuts as the dose lowers, reflecting the non-linear relationship between dose and receptor occupancy

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[34] Bogers JPAM, Hambarian G, Michiels M, Vermeulen J, de Haan L. *Risk factors for psychotic relapse after dose reduction or discontinuation of antipsychotics in patients with chronic schizophrenia: a systematic review and meta-analysis.* Schizophr Bull Open. 2020;1(1):sgaa002.

[35] Leucht S, Tardy M, Komossa K, et al. *Antipsychotic drugs versus placebo for relapse prevention in schizophrenia: a systematic review and meta-analysis.* Lancet. 2012;379(9831):2063–2071.

[36] Viguera AC, Baldessarini RJ, Hegarty JD, van Kammen DP, Tohen M. *Clinical risk following abrupt and gradual withdrawal of maintenance neuroleptic treatment.* Arch Gen Psychiatry. 1997;54(1):49–55.

— and conducted over **months**. It cannot be done on an acute ward over days, and it cannot be done at all when a patient absconds from a drug that was loaded too fast to have been tapered from safely.

What actually happened

Across **sixteen admissions, six of them following absconding**, **and at least seven agents**, the exemplar patient has **never once been tapered** by the treating team prior to discharge. Instead:

- **September 2023 — first admission:** discharged **not tapered off** and without drug-and-alcohol rehabilitation, with five benzodiazepine tablets, with immediate relapse. This first discharge set the pattern.
- **28–30 June 2026:** while on **550 mg clozapine and 10 mg aripiprazole**, the patient absconded during a walk; abrupt cessation over the following days produced **cholinergic rebound**; the hospital discharged him **in absentia**; he was conveyed by ambulance interstate in psychotic crisis.
- **On the treating team's own cited literature, his readmissions are the predicted consequence of their own practice.** Abrupt cessation → a relapse event rate of 0.77 per person-year. The crisis relied upon to justify the next detention is the crisis the withdrawal method produced.

The withdrawal-supersensitivity trap, stated plainly

Rapid loading builds the receptor supersensitivity (Part I.C). Abrupt withdrawal then unmasks it. The rebound is labelled relapse. The relapse justifies re-loading. Re-loading deepens the supersensitivity. The label escalates from "relapsing" to "treatment-resistant." Clozapine follows. **Each step is iatrogenic. None is a property of the underlying person.** The exemplar's current presentation — agitation, pressured speech, dysphoric preoccupation — is, on the evidence, cholinergic rebound from abrupt clozapine cessation compounded by benzodiazepine reduction and substance withdrawal: a withdrawal state, not proof of an illness requiring the reinstatement of the drug whose withdrawal caused it. This is not speculative. After only four weeks of clozapine, 13% of patients stopping abruptly experience moderate to severe nausea, vomiting and diarrhoea, and 40% mild symptoms; and cholinergic rebound is characterised by **agitation, fear and hallucinations, which may be mistaken for psychotic relapse** (Shiovitz et al., *Schizophrenia Bulletin* 1996, as reported in Horowitz et al. 2021)[37]. Motor withdrawal symptoms — dyskinesia, parkinsonism, neuroleptic malignant syndrome — have a reported incidence of **31–50% after abrupt first-generation antipsychotic withdrawal**, and NMS has been noted to occur on abrupt withdrawal of antipsychotics, particularly clozapine, although no systematic study of incidence has been conducted (Horowitz et al., Schizophrenia Bulletin 2021;47(4):1116–29) [38].

The evidence base for indefinite treatment is itself built on withdrawal studies

The justification for lifelong antipsychotic prescription rests on maintenance trials. But those are **discontinuation trials**: they randomise people already established on antipsychotics, often for years, to continue or to stop. The apparent benefit of continuing may therefore reflect **the harm of stopping** rather than any benefit of long-term treatment (Moncrieff, Gupta & Horowitz, *Therapeutic Advances in*

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[37] Shiovitz TM, Welke TL, Tigel PD, et al. *Cholinergic rebound and rapid onset psychosis following abrupt clozapine withdrawal.* Schizophr Bull. 1996;22(4):591–595.
[38] Horowitz, Jauhar, Natesan, Murray & Taylor, *Schizophrenia Bulletin* 2021;47(4):1116–29

Psychopharmacology 2020;10:2045125320937910)[39]. Withdrawal symptoms and withdrawal-induced psychosis are then counted as relapse — the more readily because there is **no agreed definition of relapse** across these trials, which frequently define it as a small increase across a broad range of symptoms and behaviours (Moncrieff et al., *Schizophrenia Research* 2019)[40].

The strongest contrary finding should be stated, because it is the one that will be produced against this submission. In a 20-year nationwide follow-up of first-episode schizophrenia, the risk on discontinuation rose with the duration of prior exposure: roughly doubling after one to two years, tripling after two to five, and rising sevenfold after eight (Tiihonen, Tanskanen & Taipale, *Am J Psychiatry* 2018;175(8):765–773)[41].

That finding is read as an argument for never stopping. It is more accurately an argument for never starting without a plan for stopping. What it describes is an exposure-dependent dependence that deepens with every year of treatment — a drug that becomes progressively harder to leave the longer it is given, and a risk that is created by the prescribing rather than by the illness. Read that way it is not a defence of indefinite prescription. It is the case against beginning indefinite prescription in a seventeen-year-old, and the case for the shortest exposure compatible with safety.

This was known from the beginning. Withdrawal from neuroleptics was described in *American Journal of Psychiatry* in **1959**[42], and a study of the simultaneous abrupt withdrawal of these drugs in **500 chronic psychiatric patients** was published there in **1961** (Brooks, Am J Psychiatry 1959; Judah, Josephs & Murphree, Am J Psychiatry 1961;118(2))[43]. Sixty-five years later, no withdrawal guideline exists.

The circularity is complete: the drugs create the dependence; the dependence produces the withdrawal; the withdrawal is recorded as relapse; the relapse is cited as proof the drugs are necessary.

Patients describe the loop from inside it. In the largest international survey of antipsychotic users to date — 832 people across thirty countries, predominantly USA, UK and Australia — **70% wanted to stop**, most commonly because of adverse effects and fear for their long-term physical health; **65% experienced withdrawal effects on trying**, and **51% of those described them as severe**. Of 650 free-text accounts, **57.7% reported only negative experiences** of the drugs (Read & Williams, *Current Drug Safety* 2019; Read & Sacia, *Schizophrenia Bulletin* 2020)[44]. One respondent stated the thesis of this submission in a sentense: *"Withdrawal symptoms were always blamed on relapse of my 'disease'."*

It should also be noted that this is not a class of drug confined to specialist use. Of all antipsychotic prescriptions in UK primary care between 2007 and 2011, **fewer than half of first-generation and 36–62% of second-generation recipients had any diagnosis of psychosis or bipolar disorder**, and **up to**

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[39] Moncrieff J, Gupta S, Horowitz MA. *Barriers to stopping neuroleptic (antipsychotic) treatment in people with schizophrenia, psychosis or bipolar disorder*. Ther Adv Psychopharmacol. 2020;10:2045125320937910.
[40] Moncrieff J, Crellin NE, Long MA, et al. *Definitions of relapse in trials comparing antipsychotic maintenance with discontinuation or reduction*. Schizophr Res. 2019.
[41] Tiihonen J, Tanskanen A, Taipale H. *20-year nationwide follow-up study on discontinuation of antipsychotic treatment in first-episode schizophrenia. Am J Psychiatry* 2018;175(8):765–773
[42] Brooks GW. Am J Psychiatry. 1959.
[43] Judah LN, Josephs ZM, Murphree OD. *Results of simultaneous abrupt withdrawal of ataraxics in 500 chronic psychiatric patients*. Am J Psychiatry. 1961;118(2). Also: Lacoursiere RB, Spohn HE, Thompson K. *Medical effects of abrupt neuroleptic withdrawal.* Compr Psychiatry. 1976;17(2):285–294. doi:10.1016/0010-440X(76)90002-X.
[44] Read J, Williams J. *Positive and negative effects of antipsychotic medication: an international online survey of 832 recipients.* Curr Drug Saf. 2019. Also: Read J, Sacia A. *Using open questions to understand 650 people's experiences with antipsychotic drugs.* Schizophr Bull. 2020. doi:10.1093/schbul/sbaa002.

17% had no primary mental health diagnosis at all (Marston et al., *BMJ Open* 2014)[45]. The withdrawal problem is therefore substantially larger than the schizophrenia population.

What follows is not "stopping." It is de-prescribing — a planned, supervised, documented reduction completed before discharge, not after it. No patient in this record has ever been de-prescribed. Sixteen admissions have ended in one of three ways: discharge on full dose, discharge in absentia, or absconding. On the first, at seventeen, he was handed five benzodiazepine tablets at the door and sent home. There was no taper plan, some distant appointment "after six months of stabilisation" in absence of basic metabolic and pharmacogenomic testing, and no rehabilitation. **The first thing this system did to a seventeen-year-old with a drug problem was give him a dependence-forming drug in his fist and no method of getting off it.** Every subsequent admission is downstream of that discharge.

And what precedes de-prescribing is measurement. Nothing in this submission asks that psychiatry stop treating psychosis. It asks that before a young person is committed to an indefinite exposure that becomes harder to reverse every year, somebody establishes what is actually wrong: metaboliser status and plasma concentration, so that dose means exposure; functional nutritional status rather than serum proxies — 25-hydroxyvitamin D, functional B12 by methylmalonic acid or holotranscobalamin, red cell folate, and red cell magnesium — together with homocysteine as a measure of methylation; inflammatory markers on the mandated schedule; and the substance use disorder that is the recorded primary condition. **Not one of those was done before the first prescription, and most have not been done in thirty-four months.** The proposition this submission puts to a Royal Commission is not that these drugs should never be used. It is that no one can know whether they were needed, because the question was never asked in a form capable of being answered — and that a system which does not ask it will go on producing patients like this one indefinitely, and calling the result an illness.

A urine drug screen costs a few dollars and takes two hours. It was not taken from a young man with a recorded substance use disorder who had just been run over by a bus. That is the standard of investigation against which the confidence of the diagnosis should be measured.

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[45] Marston L, Nazareth I, Petersen I, Walters K, Osborn DPJ. *Prescribing of antipsychotics in UK primary care: a cohort study.* BMJ Open. 2014;4(12).

PART IV. THE SCALE — THE MORTALITY GAP

The mortality gap is real, sourced, and partly modifiable

- People with a schizophrenia diagnosis die **15–20 years** prematurely (Correll et al., systematic review and meta-analysis, *World Psychiatry* 2022)[46].
- Across mental disorders the pooled figure is **14.66 years of potential life lost** (95% CI 13.88–15.45). Substance use disorders carry the largest gap of any diagnosis examined — **20.38 years** (18.65–22.11) — with a pooled life expectancy of **57.07 years** (54.47–59.67). Schizophrenia carries **15.22 years** (13.8–16.65) (Chan et al., eClinicalMedicine 2023;65:102294)[47]. Substance use disorder is the exemplar patient's actual primary, and untreated, condition.
- The literature identifies quality of care, second-generation antipsychotic use, and **fewer psychiatric readmissions** among the factors associated with **lower** mortality (Correll 2022; Popa A-V et al., Schizophrenia (Heidelb) 2025, 10-year cohort, 17-year observed gap)[48]. A revolving door that maximises readmissions and escalates blockade sits on the adverse side of these modifiable factors.

A comparison within that single table bears directly on this case. Substance use disorder — the exemplar patient's documented primary condition — carries a gap of 20.38 years. Schizophrenia carries 15.22. The confidence intervals do not overlap — though these are separately pooled estimates rather than a direct comparison. He has been treated intensively, across sixteen admissions and seven antipsychotic agents, for the diagnosis carrying the **smaller** gap; the condition carrying the **larger** one has received no treatment at all in thirty-four months. Pooled life expectancy for substance use disorder is 57.07 years. He is twenty.

This gap is a population figure for the diagnosed cohort. It is not, and is not offered as, a direct measure of iatrogenic causation. The honest point is narrower and sharper: the practices documented here plausibly *widen* rather than narrow a gap the literature says is partly within clinical control — and no one is measuring whether they do.

The gap has not narrowed in thirty years — and the protective claim is contested

The decisive fact is not the size of the gap but its **immobility**. The mortality gap in schizophrenia **has hardly changed in at least the last thirty years** (Tanskanen, Tiihonen & Taipale, *Acta Psychiatrica Scandinavica* 2018;138(6):492–499, 30-year nationwide follow-up)[49] — across the entire era of second-

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[46] Correll CU, Solmi M, Croatto G, Schneider LK, Rohani-Montez SC, Fairley L, Smith N, Bitter I, Gorwood P, Taipale H, Tiihonen J. *Mortality in people with schizophrenia: a systematic review and meta-analysis of relative risk and aggravating or attenuating factors*. World Psychiatry. 2022;21(2):248–271.

[47] Chan JKN, Correll CU, Wong CSM, Chu RST, Fung VSC, Wong GHS, Lei JHC, Chang WC. *Life expectancy and years of potential life lost in people with mental disorders: a systematic review and meta-analysis*. eClinicalMedicine. 2023; 65:102294. doi:10.1016/j.eclinm.2023.102294 · PMID 37965432 — 14.66 years YPLL overall; substance use disorders largest at 20.38 years; schizophrenia 15.22 years.

[48] Popa A-V, Ifteni PI, Țâbian D, Petric PS, Teodorescu A. *What is behind the 17-year life expectancy gap between individuals with schizophrenia and the general population?* Schizophrenia (Heidelb). 2025. doi:10.1038/s41537-025-00667-1 — Romanian cohort, 635 patients, 10-year follow-up; SMR 1.58; mean age at death 58.97 years; second-generation antipsychotics (HR 0.37) and low hospitalisation frequency (HR 0.09) associated with reduced mortality.

[49] Tanskanen A, Tiihonen J, Taipale H. *Mortality in schizophrenia: 30-year nationwide follow-up study*. Acta Psychiatr Scand. 2018;138(6):492–499.

generation antipsychotics, in the countries with the best data and the best-funded services. Whatever else these decades achieved, they did not move this number.

Against that, the claim that antipsychotics *reduce* mortality has been challenged in *Psychological Medicine* by David Taylor — author of the Maudsley Prescribing Guidelines — and Mark Horowitz (Taylor & Horowitz, *Psychological Medicine* 2020;50:2814–2815)[50]. Their objections are methodological, not ideological:

- The naturalistic studies classify exposure by **outpatient prescription collection**, while most deaths occur in hospital, where prescriptions are not collected — a clear route to misclassification.
- The use of **person-years** transfers deaths attributable to the cumulative effects of years of exposure into the "off-medication" group.
- On their reading of Tiihonen's own 2009 cohort, patients with **0–6 months** of antipsychotic exposure had the **lowest** mortality risk[51].
- The supporting randomised evidence has a **median duration of six weeks** (IQR 4–10) — too short for metabolic effects to manifest at all — and the meta-analysis cited as conclusive found **no significant mortality difference** in schizophrenia (OR 0.69; 95% CI 0.35–1.35; Schneider-Thoma et al., *Lancet Psychiatry* 2018)[52].
- Higher mortality in placebo arms may itself reflect **withdrawal**, prior cessation being common practice in those trials — consistent with the documented **spike in suicide attempts in the months following abrupt cessation** of atypical antipsychotics (Herings & Erkens, *Pharmacoepidemiology and Drug Safety* 2003;12(5):423–424)[53].

Their conclusion is the one this submission adopts: the concern that antipsychotic treatment is **not improving the mortality gap and may be actively contributing to it** is not answered by very short-term randomised trials and contested observational studies. *Longer-term placebo-controlled trials are required to settle it.* That two clinicians of that standing say so in that journal is the point: this is an open question inside mainstream psychiatry, not a fringe assertion.

Two specific harms sit within the gap and are dose-related: sudden cardiac death rises with antipsychotic dose (Ray et al., New England Journal of Medicine 2009;360(3):225–235), and akathisia is linked to suicide (Salem et al., Current Neuropharmacology 2017). A dose-response signal is present in precisely the exemplar's age band. In a cohort of 2,067,507 patients aged 5 to 24, antipsychotic treatment was associated with increased mortality only at doses above 100 mg chlorpromazine equivalents; no significant association was found at lower doses, or in children aged 5 to 17. The authors call for further research into the increased risk of death at higher doses in young adults (Ray WA et al., JAMA Psychiatry 2024;81(3):260–269). The exemplar patient was twenty years old; he was escalated toward approximately 500 mg of clozapine in twelve days; and, on the evidence set out in Parts I and III, he was escalated on a withdrawal-driven dopamine supersensitivity presentation — the one presentation in which the published evidence indicates stable occupancy and partial, gradual switching rather than escalation.

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[50] Taylor D, Horowitz MA. *Antipsychotics and mortality — more clarity needed.* Psychological Medicine. 2020;50:2814–2815. doi:10.1017/S0033291720004535.
[51] Tiihonen J, Lönnqvist J, Wahlbeck K, et al. *11-year follow-up of mortality in patients with schizophrenia: a population-based cohort study* (FIN11 study). Lancet. 2009;374(9690):620–627.
[52] Schneider-Thoma J, Efthimiou O, Huhn M, et al. *Second-generation antipsychotic drugs and short-term mortality: a systematic review and meta-analysis of placebo-controlled randomised controlled trials.* Lancet Psychiatry. 2018;5(8):653–663.
[53] Herings RM, Erkens JA. *Increased suicide attempt rate among patients interrupting use of atypical antipsychotics.* Pharmacoepidemiol Drug Saf. 2003;12(5):423–424.

Ray 2024 establishes the risk; the presentation makes it worse than a risk[54] [55]. Iyo holds that occupancy targets derived from normal receptor density do not transfer where density is up-regulated, and Kimura that the strategy in supersensitivity is stable occupancy with partial, gradual switching. The escalation was therefore not merely risky but directionally wrong — at speed, in the age band where dose-related mortality is documented.

A second large cohort reframes the question from safety to the balance the escalation ignored. In 102,964 patients with schizophrenia the relationship between cumulative antipsychotic exposure and mortality was U-shaped: no exposure carried the highest risk, **high** exposure the next highest, and **low-to-moderate** exposure the lowest; among older patients high exposure carried the highest risk of all. The authors frame the clinical task not as maximisation but as equilibrium — *"prescribing adequate doses that effectively treat symptoms, enhance quality of life, and enable management of side effects"* — and conclude that recommended exposure is low to moderate (Yeh L-L et al., Pharmaceuticals 2023;17(1):61)[56]. Striking that balance requires knowing what the drug is doing. In the exemplar case it was never measured. The dose was driven up one side of an equation whose other side — therapeutic effect and toxicity — was never read.

The largest cohort in this Part could not see the variable this submission is about. Yeh and colleagues followed 102,964 patients for five years and graded antipsychotic exposure by defined daily dose — that is, by milligrams dispensed. Their own final limitation is that they did not measure the amount of medication patients actually received, or their blood levels of it. The same is true of every register study in this literature. Exposure is inferred from prescriptions; prescriptions are inferred to equal drug in the body. Part I of this submission establishes that in a smoking, rapidly metabolising patient the two are uncoupled by a factor of two or more, in either direction, and that nobody looks. **The mortality literature therefore cannot distinguish a patient who received a high dose from a patient who absorbed one.** Neither can the ward. That is not a criticism of the researchers, who state the limitation plainly. It is the measure of how deep the blindness runs: the variable that determines whether a drug is treating a patient or poisoning him is absent from the individual chart, absent from the hospital audit, and absent from the national datasets on which the entire safety case for these drugs rests.

One objection must be met directly. The literature does report a mortality benefit for clozapine; the largest meta-analysis finds the greatest reduction in natural-cause mortality among clozapine-treated patients. That finding is not disputed here. But every study reporting it studies clozapine **as protocol-administered** — with mandated haematological monitoring, myocarditis screening on the recognised timeline, bowel-function review, and, in competent practice, plasma-level monitoring. The reported benefit is the benefit of a monitored regimen, not of a molecule. In the exemplar case no plasma level, and no troponin or C-reactive protein series has been produced, and myocarditis has never been excluded. **A service cannot claim the safety profile of a protocol it did not follow.**

A second, structural caution applies to that literature. Clozapine registries are survivorship-weighted by design: patients who develop agranulocytosis, myocarditis or other serious adverse reactions are withdrawn from the drug, so "current clozapine use" selects for those who tolerated it, and their deaths are counted against other exposures or against none. This is a feature of how the cohorts are assembled, not

[54] Salem H, Nagpal C, Pigott T, Teixeira AL. *Revisiting antipsychotic-induced akathisia.* Curr Neuropharmacol. 2017;15(5):789–798.
[55] Ray WA, Chung CP, Murray KT, Hall K, Stein CM. *Atypical antipsychotic drugs and the risk of sudden cardiac death.* N Engl J Med. 2009;360(3):225–235.
[56] Yeh L-L et al., Pharmaceuticals 2023;17(1):61)

an imputation of motive. It is also consistent with the finding that clozapine cessation itself marks a period of elevated suicide risk — which is the withdrawal argument of Part III, appearing in the mortality data.

What is not measured

No routine audit publishes, for these facilities, the rate of pharmacogenomic testing before escalation, the rate of supervised tapering at discharge, readmission intervals, rates of iatrogenic movement disorder, or standardised mortality. The scale of the pattern documented in this submission is therefore unknown — not because it is small, but because no one counts it. That is itself the finding, and a Royal Commission's first task is to make it knowable.

PART V. THE CASE AS EXEMPLAR — A COMPRESSED CHRONOLOGY

Every element of the systemic argument is documented in one continuous record, so far as that record has been disclosed. The exemplar patient is twenty years old and was first admitted at seventeen.

- **9 Sep 2023 — first admission.** Discharged without taper, without rehabilitation, with five benzodiazepine tablets. Immediate relapse. *The pattern is set.*
- 3 Jul 2024 — the admitting psychiatrist commences olanzapine depot with no pharmacogenomic testing (first escalation, prescribing blind), recurrent vomiting and falls over commence.
- **27 Dec 2024 – 2 Jan 2025 — Robina (Gold Coast).** He presented having absconded during transfer between the emergency department and the ward, acutely psychotic. The Guardian, then his carer, spoke with the treating doctors in the emergency department and on the ward, including one conversation of approximately an hour, supplying the collateral history, the medication history and the risks, and asking that he be transferred to Tweed rather than discharged. He was discharged on 2 January by taxi. The discharge summary records his address as "UNKOWN UNKNOWN" and his discharge destination as "Home/usual residence"; it records a telephone call to his father, who could not collect him; **it records no contact with the person who had provided the history.** "Follow Up Arrangements: Nil Entered." "Recommendations to Patient: Nil Entered." He did not arrive anywhere. Three days later he was under a bus.
- **2 Jan 2025 — serum 25-hydroxyvitamin D recorded at 68 nmol/L**, within the reference range of 50–150, on bloods taken during that admission. **He entered 2025 with normal vitamin D status, on the service's own testing.**
- **5 Jan 2025 — run over by a bus.** He collapsed in the roadway, went under the vehicle and was run over. He has never been able to say why he rolled down. He was operated on, and on 8 January admitted to the orthopaedic ward. On 8 January he was transferred to Kurrajong and remained a mental health inpatient until 2 February.

  No toxicology screen was performed on that admission. The recorded primary condition is substance use disorder; the Gold Coast record of five days earlier carries a secondary diagnosis of acute intoxication and an impression of recent substance use; he had received a 300 mg olanzapine long-acting injection six days before, and had been discharged three days before. A urine drug screen is the cheapest test in the hospital and it was not taken. **The candidate causes of the collapse are orthostatic hypotension from a recently administered depot, sedation, intoxication and seizure. Two of the four were testable on arrival. Neither was tested, and no differential appears in the record**.
- **8 Jan – 2 Feb 2025 — Kurrajong.** Approximately one month as a mental health inpatient, non-weight-bearing, with six weeks off the foot prescribed until at least the end of March. **No physiotherapy was provided and none appears in any record produced. No rehabilitation of any kind was arranged.** He was discharged on crutches, non-weight-bearing, and began weight-bearing roughly half-way through the prescribed period because he could not take care of self, relapsing into substance use on the first day of discharge.

  Throughout, the records misdescribe the discharge destination. Discharge documents across these admissions variously name addresses and persons to whom the patient was said to have been released. In every instance, apart from one, it was the Guardian who collected him, transported him, housed him and cared for him. **The service that recorded a destination did not verify one.**
- **Mar 2025 — during the Cyclone Alfred flood events**, the Guardian recovered him from the street: hopping on a moonboot in the rain, crutches lost, the limb inflamed, severely swollen and carrying an

infected laceration. He was standing ankle-deep in floodwater, barefoot but for the moonboot, its lining saturated, and was not oriented to his situation. The Guardian searched flooded roads to find him while the area was being evacuated. He was readmitted within a day or so. **This was inside the six-week non-weight-bearing period and days before the radiography the surgeons had scheduled for approximately 14 March. Radiography was performed only after weeks of demand by the Guardian. No physiotherapy and no timely orthopaedic review were provided**. Zuclopenthixol was commenced during this period. To the date of this submission — eighteen months and eleven admissions after the operation — there has been no physiotherapy.

- **Feb–Jul 2025, overlapping the above — shuffled Lismore/Tweed.** Zuclopenthixol depot commenced (second escalation) → tremor and drooling. Team proposes clozapine; patient and Guardian **refuse — recorded in the discharge summary of 15 July 2025**, which also records "poor response to multiple antipsychotic medications" and recommends drug-and-alcohol and rehabilitation **that was never provided**. ECT raised, and declined.
- **Mid-2025 — vitamin D supplementation was commenced during an inpatient admission, at a severely depleted level.** The Guardian was informed of the result at the time; the record containing it has not been produced. Supplementation is not commenced at a normal level, and was not commenced at one here. It was continued briefly, then discontinued, and no repeat level confirming correction appears in any record produced. Below approximately 25–30 nmol/L both osteomalacia and muscle weakness particularly affecting the **proximal** muscles are established, and both are reversed by treatment (Mason RS, Rybchyn MS, Brennan-Speranza TC, Fraser DR, *Faculty Reviews* 2020;9:19, University of Sydney; Girgis et al., *Endocr Rev* 2013;34(1):33–83)[57]. His recorded level was half that threshold. Myopathy of this kind is documented **without any biochemical sign of bone involvement** (Glerup et al., *Calcif Tissue Int* 2000;66(6):419–424)[58], so a normal bone profile would not have excluded it. In Australia the highest prevalence of deficiency falls in the 18–34 age band (Mason et al. 2020) — his. By mid-2025 his impaired gait and mobility had three candidate contributors running at once: iatrogenic parkinsonism, which the service itself diagnosed that July-August; proximal muscle weakness from a vitamin D deficiency it had commenced treating; and an internally-fixated lower limb never rehabilitated since the operation. No differential separating them appears anywhere. He was assessed as a psychiatric patient with a cascading adverse effects, on a leg nobody cared for.

  The interval is not mysterious. Between the normal January level and the deficiency treated later that year he was run over, operated on, hospitalised for a month, non-weight-bearing for six weeks and in a moonboot into March, then readmitted — immobilised, confined indoors and institutionally fed through an Australian autumn and winter. Depletion in those conditions is foreseeable and screenable, and a baseline existed against which any fall would have been visible. **The service held the baseline. It did not look again.**
- **Same period — lithium and haloperidol.** The discharge summary lists haloperidol among the medications administered but gives no dates for any agent, so no sequence can be established from it. It makes no mention of lithium, which the Guardian knows was administered during the same period, the patient having told her so at the time and the Guardian having observed him in a state of gross swelling, uncontrollable violent shakes, entire head skin peeling off and frozen mask-face on visiting. The discharge summary of that period records "**mild tremor at rest**". Haloperidol is a high-potency first-generation antipsychotic carrying among the highest extrapyramidal risk of any agent in common

[57] Mason RS, Rybchyn MS, Brennan-Speranza TC, Fraser DR, *Faculty Reviews* 2020;9:19, University of Sydney; Girgis et al., *Endocr Rev* 2013;34(1):33–83

[58] Glerup et al., *Calcif Tissue Int* 2000;66(6):419–424

use. **The combination of lithium carbonate with high-dose haloperidol is associated with a severe encephalopathic syndrome — lethargy, fever, tremulousness, confusion, and extrapyramidal and cerebellar dysfunction — of which two of four reported cases sustained irreversible brain damage and two were left with persistent dyskinesias** (Cohen & Cohen, JAMA 1974;230(9):1283–1287)[59]. Lithium additionally requires serum level monitoring, renal function and thyroid function testing as a condition of safe prescribing. A summary that omits an administered drug and dates none of those it lists cannot establish that any of this occurred. What cannot be established from that summary is the haloperidol dose, the dates, or whether lithium was co-administered at all — because the summary dates none of the agents it lists and does not mention the one the Guardian knows was given. That is the finding. Cohen and Cohen reported four cases and did not identify the causal factors; nobody can say from this record whether the exemplar was exposed to the same combination, at what dose, or for how long. The only party in a position to answer is the service that wrote the record, and the record it wrote cannot.

- **Aug 2025 — the treating consultant directs clozapine** (third, gravest escalation) on the assumption that readmissions = resistance, **without reading the contraindications**. Guardian objects.
- 18 Aug 2025 — Enduring Guardianship and POA. 25 Sep 2025 — discharge summary records iatrogenic Parkinsonism "secondary to the iatrogenic dopamine blockade".[60] **That entry was not volunteered. It was made only after sustained documentation and pressure by the Guardian, who held contemporaneous video and written records of the deterioration and pressed them on the service until the finding was entered on the file. The wording is the service's own.** The service's response to the harm it diagnosed: extend the dosing interval by a week and add propranolol — a second drug for the first drug's harm. Vitamin B6 was never offered. In a randomised, double-blind, placebo-controlled crossover trial of 50 inpatients — conducted in tardive dyskinesia but measured on the full Extrapyramidal Symptom Rating Scale — the parkinsonism subscale improved by −18.5 points on B6 against −1.4 on placebo ($p<0.00001$), and the dyskinesia subscale by −5.2 against −0.8 (Lerner et al., J Clin Psychiatry 2007)[61]. Magnesium was not offered either. In an animal model, magnesium given alongside a typical antipsychotic prevented the extrapyramidal effects from developing, and reduced their intensity when given after onset (Kronbauer et al., *Behav Brain Res* 2017;320:400–411 — rodent study, offered as mechanism rather than as clinical evidence)[62]. Neither B6 nor magnesium carries the adverse-effect profile of propranolol. Neither was raised. A second drug was.
- 10 Dec 2025 — NCAT appoints Guardian and Financial Manager.
- 7 Jan 2026 — testing finally done. 12 Feb 2026 — the treating consultant records the metaboliser finding (induction-prone, with clearance of olanzapine and clozapine accelerated by smoking; predictable for aripiprazole). Guardian serves an oral Cease and Desist re clozapine, acknowledged on tape. Clozapine continued and escalated. This is the graver breach: prescribing against the evidence, not merely without it.

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[59] Cohen WJ, Cohen NH. *Lithium carbonate, haloperidol, and irreversible brain damage*. JAMA. 1974;230(9):1283–1287. doi:10.1001/jama.1974.03240090023018

[60] Shin H-W, Chung SJ. *Drug-induced parkinsonism***.** J Clin Neurol 2012;8(1):15–21 — persists or progresses in 10–50% after drug withdrawal. [F]

[61] Lerner V, Miodownik C, Kaptsan A, Bersudsky Y, Libov I, Sela BA, Witztum E. *Vitamin B6 treatment for tardive dyskinesia: a randomized, double-blind, placebo-controlled, crossover study*. J Clin Psychiatry 2007;68(11):1648–1654. doi:10.4088/jcp.v68n1103 · PMID 18052557 — 50 inpatients, B6 1200 mg/day, 26-week crossover, outcome measured on the Extrapyramidal Symptom Rating Scale. [F]

[62] Kronbauer et al., *Behav Brain Res* 2017;320:400–411

- **16 Feb 2026 — clozapine administered under threat of injection**, after a nurse promised it would not be given; a cardiac complaint recorded the same day. **~by day 23 — tachycardia, atrial abnormality, chest pain, dyspnoea** (the recognised presentation of clozapine myocarditis; NSW Health GL2022_011 requires weekly troponin/CRP for four weeks)[63]. **Myocarditis has never been excluded. 18 Mar — seizure. 22 Mar — 800 mg double-dose error.** Dose later escalated to **550 mg**.
- 12 May 2026 — MHRT extends detention three months; Guardian not notified and excluded (s 76 breach, later admitted by the consultant psychiatrist, 20 May 2026). Clozapine continued.
- **23 Jun 2026 — the treating registrar concedes in writing** the patient is "clinically stable and suitable for discharge," inpatient care "no longer clinically indicated," continued detention not "consistent with the principle of least restrictive care." The private pathway (Aurora Currumbin, confirmed 28 May 2026; GP referral 7 April; insurance cleared 19 April) required only a **discharge summary — withheld for over two months.**
- 28 Jun — absconds on 550 mg clozapine + 10 mg aripiprazole; abrupt cessation → cholinergic rebound on dopamine supersensitive receptors; discharged in absentia; conveyed interstate to Logan Hospital, QLD. Complained that fluid drained from his lungs (pleural effusion is a recognised clozapine reaction requiring immediate, non-rechallenged discontinuation — Marchel et al., *BMC Res Notes* 2017)[64].
- 6 Jul — same day: the MHRT s 44 hearing is vacated because he is "outside NSW jurisdiction," while Tweed requests his transfer back into that jurisdiction. 9 Jul — transferred back over the Guardian's written objection; clozapine recommenced and titrated upward at rapid pace.
- **Throughout — sustained laxative treatment.** The prescribing record shows clozapine-induced constipation was identified and pharmacologically managed over a prolonged period while the causative agent was continued and escalated — the same pattern as propranolol for iatrogenic parkinsonism: a second drug for the first drug's harm. Clozapine-induced gastrointestinal hypomotility is a recognised reaction which may progress to severe constipation, ileus, bowel obstruction and death, and which the largest Australian and New Zealand pharmacovigilance study describes as remaining inadequately recognised (Every-Palmer & Ellis, CNS Drugs 2017;31(8):699–709). New Zealand regulatory guidance directs prescribers to ask patients about bowel movements regularly (Prescriber Update 44(2), June 2023). Benztropine, itself potently anticholinergic, was co-prescribed alongside one of the most anticholinergic antipsychotics in use. Prolonged frequent vomiting on a background of chronic constipation is the presentation of hypomotility progressing toward ileus.
- **10 Jul 2026** — the Guardian wrote to Consumer Relations and the ward requesting that the treating team and pharmacist review a list of nutritional supplements for compatibility with charted medication and permit those assessed as safe. The request cited the treating team's own precedent, having permitted supplementation on an earlier admission when zuclopenthixol produced bilateral parkinsonism, and again on readmission in August 2025. Products were offered unopened for inspection the same day. The letter recorded that the patient was then experiencing agitation and pressured speech following recent medication changes and the abrupt, unsupervised discontinuation of his previous medication. No written or oral response followed.
- **Underlying, and untreated across thirty-four months and sixteen admissions, six of them following absconding:** complex PTSD, an untreated childhood trauma, prolonged trauma-coerced

[63] NSW Health. Guideline GL2022_011: *Monitoring Clozapine-induced Myocarditis.* NSW Ministry of Health; 2022.
[64] Marchel D, Hart AL, Keefer P, Vredeveld J. *Multiorgan eosinophilic infiltration after initiation of clozapine.* BMC Res Notes. 2017;10:316.

attachment, parental alienation and coercive control, and substance use disorder. **No trauma treatment.** No NDIS Access Request lodged in over six months inpatient. Records withheld for eleven months across five statutory requests since the first formal in August 2025 — the evidentiary gap the hospital created is then relied upon to argue no alternative exists, and aggravated by a medico-legal department response that, since the complaints are on foot, the record retrieval process is affected.

He is twenty. At seventeen he trained jiu-jitsu and weighed seventy kilograms. He is now over a hundred-and-ten kilograms. On most days he cannot wake before midday, and it takes him great effort to lift himself upright. His speech is slurred and mumbled. He cannot look after himself. He avoids mirrors. He is ashamed of how he looks, and he cannot look at photographs of himself from before 2023. He was re-traumatised, over and over, without interruption. He smoked more and more, and I watched him cough his lungs out for months.

I have watched this happen across three years and sixteen admissions, and I have kept every document, because for a long time I was told that what I was seeing was his illness.

None of what is set out above is a theory to me. It is my family.

The exemplar is not an aberration in the system. It is the system, fully documented.

PART VI. THE SYSTEMIC FAILURES

Beneath the individual conduct sit structural failures a Royal Commission is the appropriate body to examine:

1. **Absence of tapering protocols — and a guideline vacuum that is deliberate.** Young adults with a recorded primary drug addiction are discharged without supervised tapers from antipsychotics and benzodiazepines, producing withdrawal and relapse — the outcome the meta-analysis predicts — which is then recorded as illness.

   This vacuum is not an oversight of the Australian system alone. In the United Kingdom, NICE **excluded antipsychotics** from its guideline on *Medicines Associated with Dependence or Withdrawal Symptoms*, despite inclusion being requested by **all four groups** at the scoping workshop and by the Royal College of Psychiatrists, the All-Party Parliamentary Group for Prescribed Drug Dependence, the International Institute for Psychiatric Drug Withdrawal, Mind, Bangor University, Pfizer and Grünenthal (Cooper, Grünwald & Horowitz, *Psychosis* 2020;12(1):89–93)[65]. The entirety of NICE's withdrawal guidance for antipsychotics, in CG178, is **two sentences**: withdraw gradually, and monitor for relapse for at least two years.

   It can be done otherwise. The **German National Guideline for Schizophrenia** (DGPPN, 2019)[66] recommends offering reduction **at any stage of the illness**, by shared decision-making; reducing **in very small steps at intervals of six to twelve weeks**; involving the patient's family and close confidants; and informing both patient and family about withdrawal phenomena — for which it provides a detailed symptom list. Australia has no equivalent national protocol. What it has is a single line in Australian Prescriber — that antipsychotics should, where possible, be stopped very slowly under close medical observation — and RANZCP clinical practice guidelines that address the duration of treatment while providing no method for ending it. Neither supplies a tapering schedule, a step size, an interval, or a withdrawal symptom list. In the exemplar case even the single line was not followed: across sixteen admissions, no taper was ever attempted.

   The submission from which the NICE material is drawn records one further point of direct application here: antipsychotics are often prescribed against the person's will under mental health legislation, which creates an even stronger duty to consider the impact of withdrawal effects when making treatment decisions. Where the State compels the exposure, the State owns the withdrawal.

2. **Absence of drug-and-alcohol rehabilitation within, or accessible from, acute care.** No structured program, no daily counselling, no separation of recovering patients from those actively using. Rehabilitation is recommended in discharge summaries and not provided.

3. **Absence of youth-specific programs.** Nothing directed at the distinct needs of a seventeen-to-twenty-year-old in a first drug-induced presentation, during the years of prefrontal maturation when high-potency dopamine blockade carries the greatest developmental risk.

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[65] Cooper RE, Grünwald LM, Horowitz M. *The case for including antipsychotics in the UK NICE guideline: Medicines associated with dependence or withdrawal symptoms.* Psychosis. 2020;12(1):89–93.

[66] Deutsche Gesellschaft für Psychiatrie und Psychotherapie, Psychosomatik und Nervenheilkunde (DGPPN). S3-Leitlinie Schizophrenie. Berlin: DGPPN; 2019. Also: Keks N, Schwartz D, Hope J. *Stopping and switching antipsychotic drugs.* Aust Prescr 2019;42(5):152–157. Also: Galletly C, Castle D, Dark F, Humberstone V, Jablensky A, Killackey E, et al. R*oyal Australian and New Zealand College of Psychiatrists clinical practice guidelines for the management of schizophrenia and related disorders*. Aust N Z J Psychiatry. 2016;50(5):410–472.

4. **Discontinuity of medical oversight — and the disappearance of the duty-holder.** Consultants and registrars rotate on approximately six-monthly cycles. The exemplar patient has been seen by more than twenty-five doctors as an inpatient across three years, four of them in the current admission alone. **The consequence is that no individual has ever owned his cumulative exposure.** Each clinician inherits a chart, a diagnosis and a dose, holds them for a few months, and hands them on. Each is accountable for the interval; none is accountable for the trajectory. Continuing the prescription requires no justification and carries no risk to the prescriber; unwinding it requires a judgement no one is in place long enough to make, and exposes the person who makes it. **The professional obligation is owed to the patient over the whole course of his treatment, and there is no one in the structure to whom that whole course belongs.** The pattern documented in this submission is not visible from inside a six-month rotation — which is why it has never been seen, and why it needs a body that can see all of it at once.

5. **Failure to test before escalating.** Pharmacogenomic metaboliser testing is cheap, common in its findings (Part I.B), and was not performed for over two years — then, once performed, disregarded. Therapeutic drug monitoring (plasma clozapine/norclozapine) is not routine, so sub-therapeutic exposure is invisible and "non-response" is misread.

6. **Failure to act on the testing performed, and failure to order the testing capable of answering the question.** Metabolic testing across three years was sporadic — vitamin D on at least one occasion, plasma vitamin levels two or three times in thirty-four months — and where it produced an abnormal result, that result was not sustained. Serum 25-hydroxyvitamin D was recorded at 68 nmol/L in January 2025, within range; supplementation was commenced at a severely depleted level later that year, discontinued. The consequence of not measuring is not only that a deficiency went untreated. It is that sudden falls since July 2024, a collapse in January 2025, a movement disorder and a progressive loss of function in June-August 2025 now have three possible explanations and no data to separate them — permanently. Diagnostic opportunity, once lost, cannot be recovered retrospectively; the window in which a level would have answered the question closed while the service declined to open it.

   The tests capable of resolving the question were never ordered. UK haematology guidance states that there is **no gold-standard test** for cobalamin deficiency, that the clinical picture is the most important factor in interpreting results, and that plasma **methylmalonic acid** is indicated as a second-line test to clarify uncertainties of underlying functional deficiency; serum **holotranscobalamin** is identified as a suitable future first-line assay (Devalia, Hamilton & Molloy, BCSH, Br J Haematol 2014;166(4):496–513)[67]. The same guidance records that serum folate reflects recent dietary intake rather than stores. In a patient with prolonged vomiting, thirty-four months of institutional diet, long-term cannabis use, an emergent movement disorder and a recorded 25-hydroxyvitamin D of 15, uncertainty as to functional deficiency plainly existed. No methylmalonic acid, no holotranscobalamin, no red cell folate and no homocysteine appear in the record. Whatever serum results exist, the second-line testing the guidance directs in exactly these circumstances was never performed.

   Nor was methylation status examined for 2 years 8 month into antipsychotics prescription. Homocysteine — which rises where B12, folate or B6 substrate is inadequate — was never measured,

[67] Devalia V, Hamilton MS, Molloy AM; *British Committee for Standards in Haematology. Guidelines for the diagnosis and treatment of cobalamin and folate disorders*. Br J Haematol. 2014;166(4):496–513. doi:10.1111/bjh.12959

in a long-term cannabis user with a documented history of prolonged vomiting and thirty-four months of institutional diet. His own genotype is wild-type at both principal MTHFR variants (677CC, 1298AA), so the commonest genetic explanation for hyperhomocysteinaemia is excluded before testing begins. That makes the measurement more informative, not less.

The pattern is consistent. Where serum is the definitive measure — vitamin D — the test was performed and the abnormal result was not sustained. Where serum is a first-line test requiring second-line confirmation — cobalamin — only the first-line test was performed, and a normal result was treated as conclusive. Where serum reflects recent intake rather than stores — folate — no red cell measure was taken. Where the cation is 99% intracellular and serum measures what remains — magnesium — no red cell measure was taken either, although erythrocyte magnesium is the measure used in published work on antipsychotic-treated schizophrenia patients (Nechifor, *Biomedicines* 2025;13:2249; Nechifor et al., *J Am Coll Nutr* 2004;23:549S–551S)[68].

Inflammatory monitoring was not merely omitted but mandated. NSW Health Guideline GL2022_011 requires weekly C-reactive protein for the first four weeks of clozapine treatment. No such series appears in any record produced.

7. **Abandonment of physical and rehabilitative care.** A patient detained under the Mental Health Act does not cease to have a body. In the exemplar case a nineteen-year-old sustained a lower limb injury requiring internal fixation with plates and screws, was thereafter an inpatient across multiple admissions in the same health district, and received no physiotherapy, no follow-up imaging and no proper orthopaedic review — at the time or since. He was discharged onto the street non-weight-bearing on crutches, and resumed weight-bearing roughly half-way through the prescribed period because no one was following-up. The consequences of unrehabilitated internal fixation are neither psychiatric nor speculative. **A service that does not measure a plasma level does not order an x-ray either without prolonged pressures from the Guardian. The blindness is not confined to pharmacology. It is a property of how these patients are seen.**

8. **Misallocation of funding.** The standing answer is that "there is no funding" for rehabilitation, tapering support, or NDIS facilitation — while funding to continue prescribing antipsychotics indefinitely is never in question. The absence of a funded community option manufactures the argument that the only "viable" discharge is the one the hospital prefers.

9. **The rehabilitation pathway did not require the drug, and the service knew it.** Asked directly on 12 February 2026 why the patient could not be referred to the Bloomfield rehabilitation unit while remaining on his existing medication, the treating consultant paused and then conceded: **"Someone can't be excluded from going to Bloomfield just because they're not on Clozapine. No, it doesn't work like that."** He then continued: *"But this is a decision I'm making."*

   Bloomfield is a public rehabilitation facility funded by the public. Access to it was not conditional on clozapine — on the treating consultant's own admission — and the Guardian had asked for that referral since August 2024. The referral was nonetheless coupled to commencement of the drug the Guardian had formally refused. **A publicly funded rehabilitation place was available on terms the patient already met, and was made contingent on a course of treatment he and his Guardian had expressly declined.** The transfer to Bloomfield was finally

[68] Nechifor, *Biomedicines* 2025;13:2249; Nechifor et al., *J Am Coll Nutr* 2004;23:549S–551S

ordered by the Mental Health Review Tribunal on 21 July 2026 — eighteen months after it was first requested, and after the drug had been administered anyway.

10. **Withholding of records as a mechanism, not an oversight.** The documents needed to establish exhaustion of options, to support a NDIS Access Request, and to allow a movement-disorder specialist to diagnose iatrogenic parkinsonism are held exclusively by the hospital and withheld. The gap is then cited against the patient. The NDIS declined access on the basis that options were "not yet exhausted"; the evidence of exhaustion is the very record being withheld.

11. **Diversion of the caring and social-work function against the family.** Funded social-work time — which exists to serve the patient — was directed instead at an NCAT application to remove the Guardian who has been present for all admissions, and at obstructing her court-ordered duty to keep the patient informed.

12. **Procedural insulation.** Hearings proceeding without notice to the designated carer (s 76); statutory determination clocks displaced by internal complaint timeframes; a jurisdictional argument used to vacate a discharge hearing and simultaneously to reclaim the patient. The order under which the patient is detained rests on an admitted breach.

13. **Absence of general health care for long-stay patients.** The exemplar patient has spent the greater part of three years as an inpatient. In that period no dental assessment or treatment has been arranged, and no general health screening beyond the sporadic bloods described above. A patient whose functional residence is an acute psychiatric ward due, whose prescription regimen carries known effects on salivation, weight and metabolic function, and who is not in a position to arrange his own appointments, has no route to ordinary medical and dental care at all. **The service assumed custody of his liberty, and despite persistent requests by Guardian did not assume custody of his health.**

None of these is a failure of an individual having a bad day. Each is a load-bearing feature of how the door revolves.

This submission does not allege intent, and does not need to. The criminal law has long recognised that culpability for killing does not require a desire to kill: the mental element of murder in this State includes **reckless indifference to human life** (*Crimes Act 1900* (NSW), s 18(1)(a)), and where the person survives, the cognate offences are reckless grievous bodily harm (s 35(2), maximum ten years) and causing grievous bodily harm by an unlawful or **negligent act or omission** (s 54).

The question this submission puts is whether a service that escalates a drug it has itself predicted will not work, in a patient it knows will interrupt it, without adequate risk assessment, without informed consent, without the mandated cardiac monitoring, and over a formal Cease and Desist, has crossed from negligence into reckless indifference — and, if it has, how many other young people are inside that same practice today, or have been harmed in the past. That is a question for a Royal Commission. It is not one a health district can answer about itself.

PART VII. TERMS OF REFERENCE SOUGHT

A Royal Commission (or, in the alternative, a Special Commission of Inquiry under the *Special Commissions of Inquiry Act 1983* (NSW), and a parliamentary inquiry) is asked to examine and report on:

1. **The TRS classification.** How often the label "treatment-resistant schizophrenia" is applied without documented adequacy of prior trials by **dose, duration, and plasma concentration**; and the rate at which **pseudo-resistance** (sub-therapeutic exposure) is distinguished from true resistance before clozapine is commenced.

2. **Pharmacogenomic and therapeutic-drug-monitoring practice.** The rate of metaboliser testing and plasma-level monitoring **before** dose escalation across NSW public mental health services, and the clinical governance that permits escalation against a recorded metaboliser finding.

3. **Nutritional, metabolic and inflammatory assessment.** The rate at which nutritional and metabolic status is assessed in patients receiving long-term antipsychotic treatment; whether functional markers capable of establishing tissue status are used, or only serum assays known not to reflect it; the rate at which identified deficiencies are corrected and correction confirmed; and compliance with mandated inflammatory monitoring under NSW Health Guideline GL2022_011.

4. **Titration and de-titration.** Prevailing antipsychotic **loading** rates against published titration ceilings, and the availability and use of supervised **hyperbolic tapering** at discharge; the proportion of discharges effected with **no taper**. It should further examine whether services differentiate titration strategy where dopamine supersensitivity is present, given that receptor up-regulation alters the relationship between dose, occupancy and clinical effect.

5. **Dopamine supersensitivity.** The extent to which relapse and readmission are attributable to supersensitivity generated by prior blockade rather than to an intractable underlying illness, and whether services distinguish the two before escalating. It should further examine the frequency with which acquired tolerance — non-response to successive dose increases — is recorded as evidence of treatment resistance rather than assessed as a diagnostic criterion for supersensitivity.

6. **Outcomes audit.** For each acute unit: readmission intervals, standardised mortality, rates of iatrogenic movement disorders, forced-treatment incidents, and discharges-in-absentia — published, and benchmarked, for the past twenty years, with specific reference to Tweed Valley Hospital, Lismore Base Hospital, and Byron Central Hospital. **It should further examine, for deaths occurring during or within twelve months of antipsychotic treatment, the cause of death as recorded, whether any treatment-related contribution was considered, and whether the datasets held by NSW Health are capable of establishing one.**

7. **Consent, guardianship and procedural fairness.** Compliance with designated-carer notification (s 76), the interaction of internal complaint timeframes with statutory determination clocks, and the use of jurisdictional argument to defeat discharge review.

8. **Service gaps.** The absence of integrated drug-and-alcohol rehabilitation, youth programs, tapering support, NDIS facilitation from acute care, and the absence of physiotherapy, rehabilitation and orthopaedic or other specialist physical follow-up for injuries sustained by mental health inpatients, and the funding architecture that sustains indefinite medication as the default.

9. **Records access.** The systemic withholding of health records from patients and their guardians, and its role in defeating access to alternatives, to specialist diagnosis, and to disability support.

Immediate measures urged, pending inquiry

- Mandatory pharmacogenomic testing and plasma-level monitoring before any escalation to clozapine/olanzapine.
- A prohibition on discharge (including discharge in absentia) without a documented supervised taper plan.
- Mandatory exclusion of clozapine-induced myocarditis (troponin, CRP, echocardiography) on the recognised timeline, and no rechallenge after a serositis reaction.
- Independent movement-disorder review, including DaTscan and UPDRS, where iatrogenic parkinsonism is recorded.
- Independent third-party psychiatric assessment.
- Independent orthopaedic review with current imaging where internal fixation has been performed and no rehabilitation care is received.
- Enforcement of designated-carer notification and of guardians' access to records.

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30 July 2026

This document is provided as a matter of public record and as the consolidated systemic basis for complaint to the NSW Ombudsman, the Health Care Complaints Commission (HCCC), Australian Health Practitioner Regulation Agency (AHPRA), NSW Civil and Administrative Tribunal (NCAT), the Australian Human Rights Commission (AHRC), and any Royal Commission or parliamentary inquiry into these matters. It may be cited with attribution to the author. Contact: juliaboon@me.com

REFERENCES

All references are drawn from the peer-reviewed literature or from primary legal instruments. Citations already relied upon in filed proceedings (MHRT 15 July 2026; HCCC HC00532615) are marked [F].

"Treatment resistance" vs. pseudo-resistance; pharmacogenomics

- Bondolfi G, Morel F, Crettol S, Rachid F, Baumann P, Eap CB. *Increased clozapine plasma concentrations and side effects induced by smoking cessation in 2 CYP1A2 genotyped patients.* Ther Drug Monit 2005;27(4):539–543. PMID 16044115 — threefold rise following smoking cessation.
- Cipolla et al. *Combination of Two Long-Acting Antipsychotics in Schizophrenia Spectrum Disorders: A Systematic Review.* Brain Sciences 2024. [F]
- Haslemo T et al. *Eur J Clin Pharmacol* 2006;62(12):1049–53 — smoking reduces plasma clozapine ~50%, olanzapine ~67%. [F]
- Howes OD, McCutcheon R, Agid O, de Bartolomeis A, van Beveren NJM, Birnbaum ML, et al. *Treatment-Resistant Schizophrenia: Treatment Response and Resistance in Psychosis (TRRIP) Working Group Consensus Guidelines on Diagnosis and Terminology.* Am J Psychiatry 2017;174(3):216–229. doi:10.1176/appi.ajp.2016.16050503 — at least two failed trials, each of minimum six weeks, standardised rating scales, and explicit criteria for assured adherence; non-response defined as <20% symptom improvement.
- Koonrungsesomboon N, Khatsri R, Wongchompoo P, Teekachunhatean S. *The impact of genetic polymorphisms on CYP1A2 activity in humans: a systematic review and meta-analysis.* The Pharmacogenomics Journal 2018;18(6):760–768 — n=3,570; heterozygous −163C>A carriers SMD 0.32 (p=0.003), rising to 0.56 among smokers (p=0.001); no other CYP1A2 polymorphism significant.
- Primary genotype data: AncestryDNA V2.0 array, human reference build 37, forward (+) strand, collected 3 November 2023 — rs762551 = A/C (heterozygous CYP1A2*1F); rs1801133 = G/G and rs1801131 = T/T (MTHFR 677CC and 1298AA, homozygous reference; note that the C677T and A1298C designations are cDNA nomenclature on the minus strand, so the forward-strand variant alleles are A and G respectively, neither present). Consumer array data, offered as corroborative only; the treating team's clinical pharmacogenomic report disclosed on 12 February 2026 remains the primary evidence.
- Sachse C, Brockmöller J, Bauer S, Roots I. *Functional significance of a C→A polymorphism in intron 1 of the cytochrome P450 CYP1A2 gene tested with caffeine.* Br J Clin Pharmacol 1999 Apr;47(4):445–449. doi:10.1046/j.1365-2125.1999.00898.x · PMID 10233211 — no significant genotype effect in non-smokers; 1.6-fold higher activity in A/A smokers.
- Varney L, Sanchez-Roige S, Palmer AA, et al. *Effect of CYP1A2, CYP2D6 and CYP3A4 variation on antipsychotic treatment outcomes.* Pharmaceuticals 2025;18(6):892 — n=453; 89.6% carried the inducible CYP1A2 allele; 62.9% an allele altering CYP2D6/CYP3A4 metabolism.

Dopamine supersensitivity

- Chouinard G, Samaha A-N, Chouinard V-A, Peretti C-S, Kanahara N, Takase M, Iyo M. *Antipsychotic-induced dopamine supersensitivity psychosis: pharmacology, criteria, and therapy. Psychotherapy and Psychosomatics* 2017;86(4):189–219 — reported prevalence ~30% in schizophrenia, ~70% in TRS. [F]
- Horowitz MA, Jauhar S, Natesan S, Murray RM, Taylor D. *A Method for Tapering Antipsychotic Treatment That May Minimize the Risk of Relapse.* Schizophrenia Bulletin 2021;47(4):1116–29. [F]
- Iyo M, Tadokoro S, Kanahara N, Hashimoto T, Niitsu T, Watanabe H, Hashimoto K. *Optimal extent of dopamine D2 receptor occupancy by antipsychotics for treatment of dopamine supersensitivity psychosis and late-onset*

psychosis. J Clin Psychopharmacol 2013;33(3):398–404. doi:10.1097/JCP.0b013e31828ea95c · PMID 23609386 — standard occupancy targets are of limited application where D2 density is up-regulated.

- Kimura H, Kanahara N, Sasaki T, et al. *Risperidone long-acting injectable in the treatment of treatment-resistant schizophrenia with dopamine supersensitivity psychosis: results of a 2-year prospective study.* J Psychopharmacol 2016. doi:10.1177/0269881116655978 — 108 patients, 72 with DSP history; strategy of stable D2 occupancy, oral agents switched partially and gradually.
- Lugg W. *Antipsychotic-induced supersensitivity — a reappraisal.* Aust N Z J Psychiatry 2022;56(5):437–444. PMID 34144649 — argues that chronic antipsychotic exposure induces neuroadaptive changes rendering some individuals more vulnerable to psychotic relapse, **including those receiving continuous treatment**; and that in treating every episode with prolonged or indefinite therapy the profession may paradoxically increase relapse risk in a significant proportion of people. Published in the journal of the Royal Australian and New Zealand College of Psychiatrists.

Tapering, withdrawal and relapse

- Bogers JPAM, Hambarian G, Michiels M, Vermeulen J, de Haan L. *Risk factors for psychotic relapse after dose reduction or discontinuation of antipsychotics in patients with chronic schizophrenia: a systematic review and meta-analysis.* Schizophr Bull Open 2020;1(1):sgaa002 — 40 publications, 46 cohorts, 1,677 patients; pooled relapse event rate 0.55 (95% CI 0.46–0.65) per person-year. By duration of dose reduction: abrupt 0.77 (0.56–0.98); 1–2 weeks 0.57 (0.35–0.80); 3–10 weeks 0.47 (0.28–0.67); more than 10 weeks 0.31 (0.26–0.36); P < .0001. By endpoint: abrupt discontinuation 0.77 (0.56–0.98); gradual discontinuation 0.92 (0.58–1.26); gradual reduction to a maintained rest dose 0.35 (0.25–0.44).
- Brooks GW. Am J Psychiatry 1959.
- Carbon M, Kane JM, Leucht S, Correll CU. *Tardive dyskinesia risk with first- and second-generation antipsychotics in comparative randomized controlled trials: a meta-analysis.* World Psychiatry 2018;17(3):330–340.
- Cooper RE, Grünwald LM, Horowitz M. *The case for including antipsychotics in the UK NICE guideline. Psychosis* 2020;12(1):89–93 — NICE exclusion; German DGPPN guideline as the comparator.
- Galletly C, Castle D, Dark F, Humberstone V, Jablensky A, Killackey E, et al. *Royal Australian and New Zealand College of Psychiatrists clinical practice guidelines for the management of schizophrenia and related disorders.* Aust N Z J Psychiatry 2016;50(5):410–472.
- Horowitz MA, Murray RM, Taylor D. *Tapering antipsychotic treatment. JAMA Psychiatry* 2020;77(11):1181–1182. doi:10.1001/jamapsychiatry.2020.2166 — hyperbolic tapering; final doses may need to be as little as 2.5% of a therapeutic dose.
- Judah LN, Josephs ZM, Murphree OD. *Results of simultaneous abrupt withdrawal of ataraxics in 500 chronic psychiatric patients.* Am J Psychiatry 1961;118(2).
- Lacoursiere RB, Spohn HE, Thompson K. *Medical effects of abrupt neuroleptic withdrawal.* Compr Psychiatry 1976;17(2):285–294. doi:10.1016/0010-440X(76)90002-X
- Leucht S, Tardy M, Komossa K, et al. *Antipsychotic drugs versus placebo for relapse prevention in schizophrenia. Lancet* 2012;379(9831):2063–2071 — four-week "gradual" discontinuation is no better than abrupt.
- Marston L, Nazareth I, Petersen I, Walters K, Osborn DPJ. *Prescribing of antipsychotics in UK primary care: a cohort study.* BMJ Open 2014;4(12).
- Moncrieff J, Crellin NE, Long MA, et al. *Definitions of relapse in trials comparing antipsychotic maintenance with discontinuation or reduction.* Schizophr Res 2019 — no agreed definition of relapse.
- Moncrieff J, Gupta S, Horowitz MA. *Barriers to stopping neuroleptic (antipsychotic) treatment in people with schizophrenia, psychosis or bipolar disorder.* Ther Adv Psychopharmacol 2020;10:2045125320937910 — the maintenance evidence base is built on discontinuation trials confounded by withdrawal.

- Read J, Williams J. *Positive and negative effects of antipsychotic medication: an international online survey of 832 recipients*. Current Drug Safety 2019;14(3):173–181. doi:10.2174/1574886314666190301152734 · PMID 30827259 — n=832, 30 countries, predominantly USA, UK and Australia.
- Read J, Sacia A. Using open questions to understand 650 people's experiences with antipsychotic drugs. Schizophr Bull 2020. doi:10.1093/schbul/sbaa002.
- Shiovitz TM, Welke TL, Tigel PD, et al. *Cholinergic rebound and rapid onset psychosis following abrupt clozapine withdrawal.* Schizophr Bull 1996;22(4):591–595.
- Keks N, Schwartz D, Hope J. *Stopping and switching antipsychotic drugs*. Australian Prescriber. 2019;42(5):152–157 — antipsychotics should generally be reduced and stopped slowly, over weeks to months; abrupt discontinuation risks withdrawal, relapse and rebound.
- Tadokoro S, Okamura N, Sekine Y, Kanahara N, Hashimoto K, Iyo M. *Chronic treatment with aripiprazole prevents development of dopamine supersensitivity and potentially supersensitivity psychosis.* Schizophr Bull 2012;38(5):1012–1020.
- Tiihonen J, Tanskanen A, Taipale H. *20-year nationwide follow-up study on discontinuation of antipsychotic treatment in first-episode schizophrenia. Am J Psychiatry* 2018;175(8):765–773 — relapse risk on discontinuation doubles after 1–2 years of exposure, triples after 2–5, and rises sevenfold after 8.
- Viguera AC, Baldessarini RJ, Hegarty JD, van Kammen DP, Tohen M. *Clinical risk following abrupt and gradual withdrawal of maintenance neuroleptic treatment. Arch Gen Psychiatry* 1997;54(1):49–55 — tapering over 3–9 months halves relapse; 60% of all relapses over 4 years occur within 3 months of cessation.

Clozapine titration and adverse reactions

- Devalia V, Hamilton MS, Molloy AM; British Committee for Standards in Haematology. *Guidelines for the diagnosis and treatment of cobalamin and folate disorders.* Br J Haematol 2014;166(4):496–513. doi:10.1111/bjh.12959
- Every-Palmer S, Ellis PM. *Clozapine-induced gastrointestinal hypomotility*. CNS Drugs 2017;31(8):699–709.
- Faber MS, Fuhr U. *Time response of cytochrome P450 1A2 activity on cessation of heavy smoking.* Clin Pharmacol Ther 2004;76(2):178–184.
- Government of South Australia, Department for Health and Wellbeing. *Factors influencing clozapine plasma concentrations.* Last updated March 2023 — TDM indications; smoking induction up to 50%; NRT does not induce; 7–12 cigarettes/day sufficient; caffeine inhibition; target range 350–600 μg/L; **monitoring required after smoking cessation because levels may rise 50% and cause seizures**.
- Haidary HA, Padhy RK. *Clozapine***.** In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; last updated 10 November 2023.
- Lenich A-K, Ruez S. *Evaluating CYP enzyme induction by cigarette smoking: a new in vitro approach using primary human hepatocytes*. Archives of Toxicology 2025;99:4513–4530. doi:10.1007/s00204-025-04135-8 — concentration-dependent CYP1A1/CYP1A2 induction by cigarette smoke extract in primary human hepatocytes.
- Marchel D, Hart AL, Keefer P, Vredeveld J. *Multiorgan eosinophilic infiltration after initiation of clozapine.* BMC Res Notes 2017;10:316. [F]
- Mouaffak F, Gaillard R, Burgess E, Zaki H, Olié JP, Krebs M-O. *Clozapine-induced serositis: review of its clinical features, pathophysiology and management strategies***.** Clin Neuropharmacol 2009;32(4):219–223. doi:10.1097/wnf.0b013e318197a2f2 · PMID 19620851 — polyserositis; discontinue immediately and do not rechallenge. [F]
- New Zealand Medicines and Medical Devices Safety Authority. *Antipsychotic-induced constipation — high impact for patients.* Prescriber Update 2023;44(2):24–26 — clozapine carries a high risk of constipation

and related complications which can be fatal; prescribers directed to ask patients about bowel movements regularly.

- NSW Health Guideline **GL2022_011** — *Monitoring Clozapine-induced Myocarditis*; weekly troponin/CRP for first four weeks. [F]
- Ronaldson KJ, Fitzgerald PB, Taylor AJ, Topliss DJ, McNeil JJ. *A new monitoring protocol for clozapine-induced myocarditis based on an analysis of 75 cases and 94 controls***.** Aust N Z J Psychiatry 2011;45(6):458–465 — onset 10–33 days, 83% between days 14 and 21; nine deaths; protocol recommends baseline troponin I/T, CRP and echocardiography, then troponin and CRP on days 7, 14, 21 and 28. [F]
- *Special Report: Clozapine Revisited. Psychiatric News* 2025 (APA) — standard initiation 12.5 mg BID, target 300–450 mg, TDM, ancestry-adjusted titration.

Smoking, substitution and CYP1A2 induction

- Bak JH, Lee SM, Lim HB. *Safety assessment of mainstream smoke of the herbal cigarette.* Toxicological Research 2015;31(1):41–48. doi:10.5487/TR.2015.31.1.041 · PMID 25874032 — carbon monoxide and benzo(α)pyrene higher in the herbal cigarette than in a standard cigarette of equal tar.
- Campbell JM, Lindsey AJ. *Polycyclic hydrocarbons extracted from tobacco: the effect upon total quantities found in smoking.* Br J Cancer 1956 Dec;10(4):649–652. doi:10.1038/bjc.1956.78 · PMID 13426376 — PAHs are largely formed by combustion rather than released from unburnt material.
- Cormac I, Brown A, Creasey S, Ferriter M, Huckstep B. *A retrospective evaluation of the impact of total smoking cessation on psychiatric inpatients taking clozapine.* Acta Psychiatr Scand 2010 May;121(5):393–397. doi:10.1111/j.1600-0447.2009.01482.x · PMID 19824991 — plasma levels ≥1000 μg/L rose from 4.2% to 41.7% within six months of a ward smoking ban, despite dose reductions.
- Gee SH, Taylor DM, Shergill SS, Flanagan R, MacCabe JH. *Effects of a smoking ban on clozapine plasma concentrations in a nonsecure psychiatric unit.* Ther Adv Psychopharmacol 2017;7(2):79–83. doi:10.1177/2045125316677027 — 31 patients, mean clozapine dose 502 mg/day. No significant change in clozapine or norclozapine plasma concentrations after a hospital-grounds smoking ban, and no change in reported tobacco consumption. The authors' conclusion: the ban had little impact on plasma concentrations **because patients continued to smoke tobacco if allowed to leave**.
- Gilbert JAS, Lindsey AJ. *Polycyclic hydrocarbons in tobacco smoke: pipe smoking experiments***.** Br J Cancer 1956 Dec;10(4):646–648. doi:10.1038/bjc.1956.77 · PMID 13426375 — herbal smoking mixtures yield higher proportions of polycyclic aromatic hydrocarbons than cigarette smoke.
- Mould S, Hogan R. *Guidance on the effects of smoking and smoking cessation on psychotropic and other medications.* Oxford Health NHS Foundation Trust, Drugs & Therapeutics Group; approved December 2023 — unplanned cessation on admission; re-titration target ~30% lower with plasma monitoring; enzyme saturation; record filtered vs unfiltered.
- Puljević C, Coomber R, Kinner SA, de Andrade D, Mitchell C, White A, Cresswell SL, Bowman J. *'Teabacco': smoking of nicotine-infused tea as an unintended consequence of prison smoking bans.* Drug and Alcohol Review 2018;37(7). doi:10.1111/dar.12848 — 82 people released from Queensland prisons; 57% had smoked teabacco while incarcerated.
- Zullino DF, Delessert D, Eap CB, Preisig M, Baumann P. *Tobacco and cannabis smoking cessation can lead to intoxication with clozapine or olanzapine.* Int Clin Psychopharmacol 2002 May;17(3):141–143. doi:10.1097/00004850-200205000-00008 · PMID 11981356.

Drug-induced parkinsonism, movement disorder and encephalopathy

- Cohen WJ, Cohen NH. *Lithium Carbonate, Haloperidol, and Irreversible Brain Damage***.** JAMA 1974;230(9):1283–1287. doi:10.1001/jama.1974.03240090023018 — four patients with mania treated with lithium carbonate and high-dose haloperidol developed severe encephalopathic syndromes; two sustained irreversible brain damage, two were left with persistent dyskinesias. Causal factors were not identified.

- Girgis CM, Clifton-Bligh RJ, Hamrick MW, et al. *The roles of vitamin D in skeletal muscle: form, function, and metabolism.* Endocr Rev 2013;34(1):33–83.
- Glerup H, Mikkelsen K, Poulsen L, et al. *Hypovitaminosis D myopathy without biochemical signs of osteomalacic bone involvement.* Calcif Tissue Int 2000;66(6):419–424.
- Kronbauer M, Metz VG, Roversi K, et al. *Influence of magnesium supplementation on movement side effects related to typical antipsychotic treatment in rats.* Behav Brain Res 2017;320:400–411.
- Lerner V, Miodownik C, Kaptsan A, Bersudsky Y, Libov I, Sela BA, Witztum E. *Vitamin B6 treatment for tardive dyskinesia: a randomized, double-blind, placebo-controlled, crossover study.* J Clin Psychiatry 2007;68(11):1648–1654. doi:10.4088/jcp.v68n1103 · PMID 18052557 — 50 inpatients, B6 1200 mg/day, 26-week crossover, outcome measured on the Extrapyramidal Symptom Rating Scale. [F]
- Mason RS, Rybchyn MS, Brennan-Speranza TC, Fraser DR. *Is it reasonable to ignore vitamin D status for musculoskeletal health?* Faculty Reviews 2020;9:19. doi:10.12703/r/9-19 · PMID 33659951.
- Nechifor M. *Magnesium and Zinc in Schizophrenia.* Biomedicines 2025;13:2249. doi:10.3390/biomedicines13092249 — 99% of body magnesium is intracellular.
- Nechifor M, Vaideanu C, Palamaru I, Borza C, Mindreci I. *The influence of some antipsychotics on erythrocyte magnesium and plasma magnesium, calcium, copper and zinc in patients with paranoid schizophrenia.* J Am Coll Nutr 2004;23:549S–551S.
- Shin H-W, Chung SJ. *Drug-induced parkinsonism***.** J Clin Neurol 2012;8(1):15–21 — persists or progresses in 10–50% after drug withdrawal. [F]

Mortality gap

- Buhagiar K, Templeton G, Blyth H, Dey M, Giacco D. *Mortality risk from long-term treatment with antipsychotic polypharmacy versus monotherapy among adults with serious mental illness: a systematic review and meta-analysis of observational studies.* Schizophr Res 2020. doi:10.1016/j.schres.2020.08.026 — 12 studies, 834,534 person-years; **no difference** in all-cause mortality between polypharmacy and monotherapy (crude rate ratio 0.94, 95% CI 0.81–1.10, p=0.446; adjusted HR 0.98, 95% CI 0.80–1.19, p=0.802). The paper records that practice guidelines nonetheless advise against long-term polypharmacy, on efficacy rather than mortality grounds.
- Chan JKN, Correll CU, Wong CSM, Chu RST, Fung VSC, Wong GHS, Lei JHC, Chang WC. *Life expectancy and years of potential life lost in people with mental disorders: a systematic review and meta-analysis.* eClinicalMedicine 2023;65:102294. doi:10.1016/j.eclinm.2023.102294 · PMID 37965432
- Correll CU, Solmi M, Croatto G, Schneider LK, Rohani-Montez SC, Fairley L, Smith N, Bitter I, Gorwood P, Taipale H, Tiihonen J. *Mortality in people with schizophrenia: a systematic review and meta-analysis of relative risk and aggravating or attenuating factors.* World Psychiatry 2022;21(2):248–271 — 15–20-year gap; modifiable factors include quality of care, antipsychotic use and readmissions.
- Herings RM, Erkens JA. *Increased suicide attempt rate among patients interrupting use of atypical antipsychotics.* Pharmacoepidemiol Drug Saf 2003;12(5):423–424.
- Popa A-V, Ifteni PI, Țâbian D, Petric PS, Teodorescu A. *What is behind the 17-year life expectancy gap between individuals with schizophrenia and the general population?* Schizophrenia (Heidelb) 2025. doi:10.1038/s41537-025-00667-1 — Romanian cohort, 635 patients, 10-year follow-up; SMR 1.58; mean age at death 58.97 years; second-generation antipsychotics (HR 0.37) and low hospitalisation frequency (HR 0.09) associated with reduced mortality.

- Ray WA, Chung CP, Murray KT, Hall K, Stein CM. *Atypical antipsychotic drugs and the risk of sudden cardiac death. N Engl J Med* 2009;360(3):225–235 — dose-related.
- Ray WA, Fuchs DC, Olfson M, et al. *Antipsychotic medications and mortality in children and young adults.* JAMA Psychiatry 2024;81(3):260–269 — cohort of 2,067,507 patients aged 5–24; increased mortality only at doses above 100 mg chlorpromazine equivalents; no significant association at lower doses or in children aged 5–17.
- Salem H, Nagpal C, Pigott T, Teixeira AL. *Revisiting antipsychotic-induced akathisia. Curr Neuropharmacol* 2017;15(5):789–798 — akathisia-linked suicide.
- Schneider-Thoma J, Efthimiou O, Huhn M, et al. *Second-generation antipsychotic drugs and short-term mortality. Lancet Psychiatry* 2018;5(8):653–663 — no significant mortality difference in schizophrenia (OR 0.69; 95% CI 0.35–1.35); median study duration 6 weeks.
- Tanskanen A, Tiihonen J, Taipale H. *Mortality in schizophrenia: 30-year nationwide follow-up study. Acta Psychiatr Scand* 2018;138(6):492–499 — the gap has hardly changed in thirty years.
- Taylor D, Horowitz MA. *Antipsychotics and mortality — more clarity needed. Psychological Medicine* 2020;50:2814–2815. doi:10.1017/S0033291720004535 — methodological critique of the claim that antipsychotics reduce mortality.
- Tiihonen J, Lönnqvist J, Wahlbeck K, et al. *11-year follow-up of mortality in patients with schizophrenia: a population-based cohort study (FIN11 study).* Lancet 2009;374(9690):620–627 — note: FIN11's own conclusion was that longer cumulative antipsychotic exposure was associated with lower mortality (HR 0.81, 95% CI 0.77–0.84); the contrary sub-finding cited in Part IV is Taylor and Horowitz's reading of it.
- Yeh L-L, Lee W-C, Kuo K-H, Pan Y-J. *Antipsychotics and mortality in adult and geriatric patients with schizophrenia.* Pharmaceuticals (Basel) 2023;17(1):61. doi:10.3390/ph17010061 — Taiwan NHIRD, 102,964 patients; U-shaped exposure–mortality relationship. Non-exposure carried the highest risk (3.61-fold overall, 3.37-fold cardiovascular), followed by high exposure; low to moderate exposure carried the lowest. Among older patients high exposure carried the highest risk (3.01-fold overall, 2.95-fold cardiovascular). The authors conclude that recommended exposure is low to moderate.

Legal instruments

- *Health Care Complaints Act 1993* (NSW); Health Records and Information Privacy Act 2002 (NSW).
- *Mental Health Act 2007* (NSW), ss 12, 42, 43, 44, 76.
- *Rogers v Whitaker* (1992) 175 CLR 479; *Secretary, DHCS v JWB and SMB* (Marion's Case) (1992) 175 CLR 218; *Civil Liability Act 2002* (NSW), ss 5B, 5D, 5O; *Northern Territory v Mengel* (1995) 185 CLR 307.